



How to decide Q-Point?

- If transistor is to be used as a switch, then Q-point must lie in cut off or saturation region.

Saturation: ON switch

cut off : OFF switch

- If transistor is to be used as an amplifier, the Q-point must be in linear region.





How to decide Q-Point?

- If transistor is to be used as a switch, then Q-point must lie in cut off or saturation region.

Saturation: ON switch

cut off : OFF switch

- If transistor is to be used as an amplifier, Q-point must be in

↳ generally in active region we bias T_x and line

distant from
& cut off



How to decide Q-Point?

- If transistor is to be used as a switch, then Q-point must lie in cut off or saturation region.

Saturation: ON switch

cut off : OFF switch

- If transistor is to be used as an amplifier, then Q-point must be in Active Region.

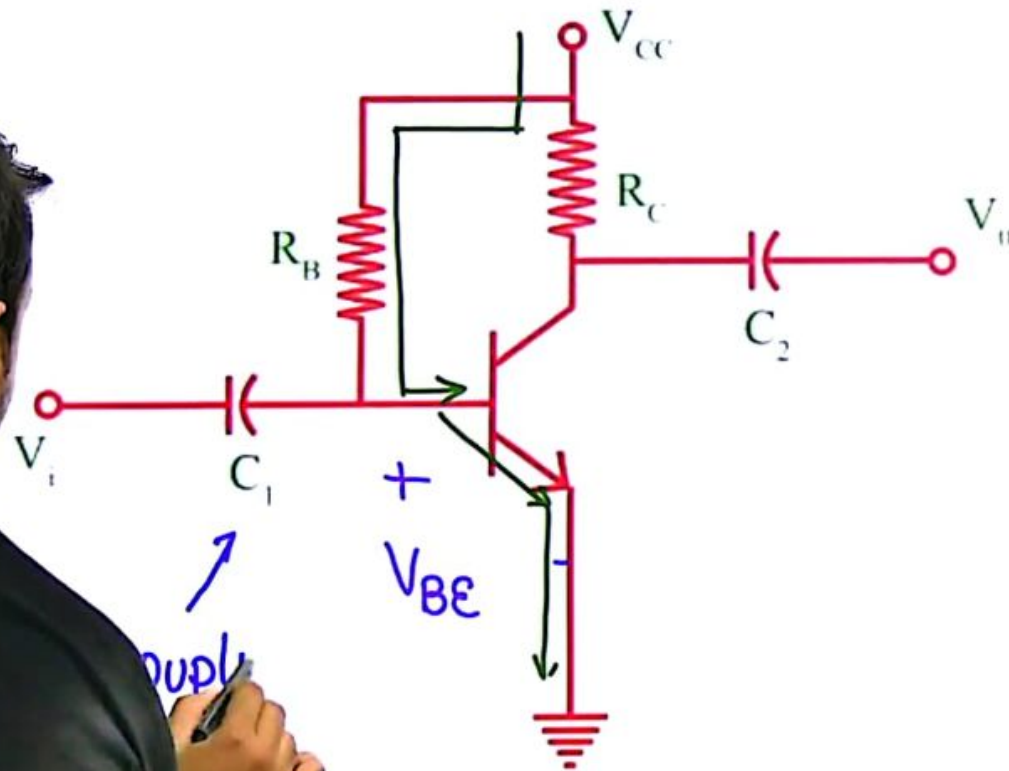
↳ generally in active region we bias T_x at mid of load line

↳ equidistant from sat & cut off

↳ max^m gain or amplification



FIXED BIAS CONFIGURATION



• we directly connect a voltage source at base terminal.

$$V_{IN} = \text{fixed}$$

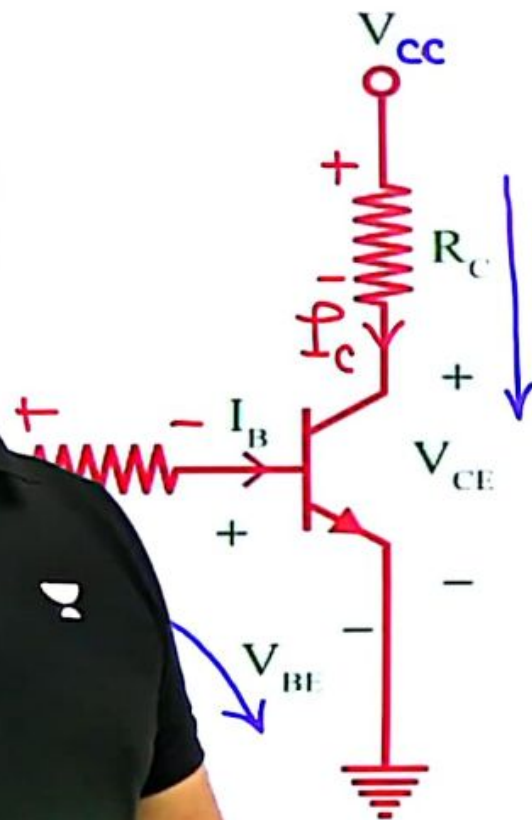
dc analysis

$$\omega = 0$$

$\frac{1}{\omega C} \rightarrow \infty$, capacitor: open



At only DC signal



$$\rightarrow V_{CC} - I_B R_B - V_{BE} = 0 \quad (i)$$

$$\rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\rightarrow I_C = \beta I_B$$

$$\rightarrow V_{CC} - I_C R_C - V_{CE} = 0$$

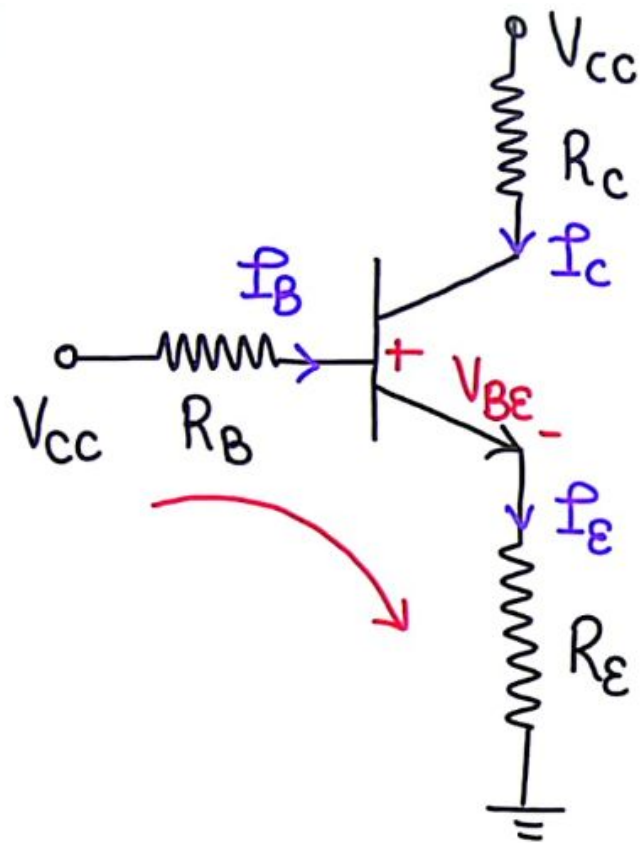
$$V_{CE} = V_{CC} - I_C R_C \begin{cases} > 0.2 \text{ active} \\ \leq 0.2 \text{ sat} \end{cases}$$

diff (i) wrt I_C

$$0 - \frac{\partial I_B}{\partial I_C} R_B - 0 = 0 ; \frac{\partial I_B}{\partial I_C} = 0$$

$$S = 1 + \beta = \text{high}$$

$$S = \frac{1 + \beta}{1 - \beta \frac{\partial I_B}{\partial I_C}}$$



$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - I_B R_B - V_{BE} - (I_C + I_B) R_E = 0$$

diff w.r.t I_C

$$0 - R_B \frac{\partial I_B}{\partial I_C} - 0 - R_E = 0$$

$$\frac{\partial I_B}{\partial I_C}$$

more

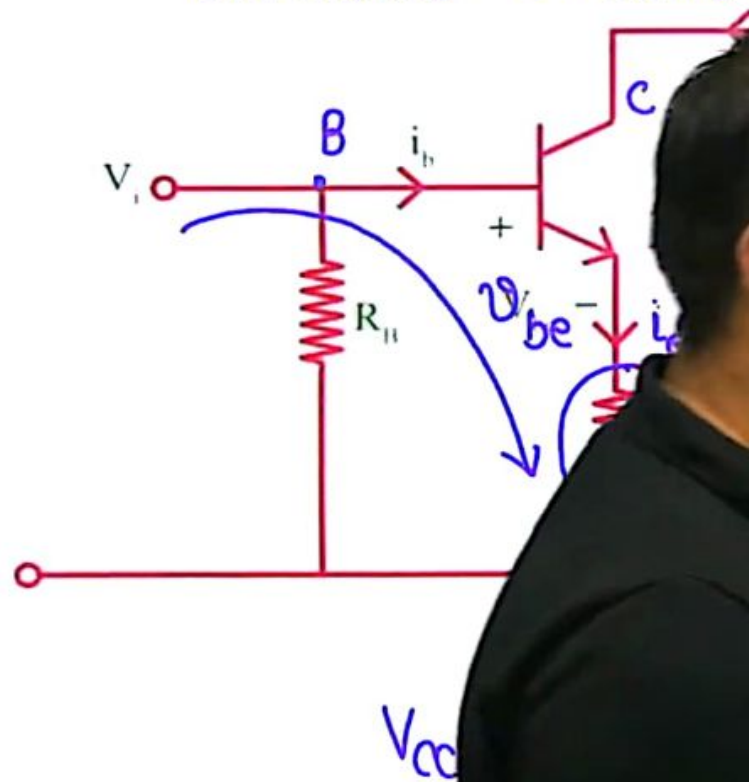
$\leq 1 + \beta$: s'

B



Drawback of Emitter Biasing

ac model : $C \rightarrow \text{short}$



$$v_{be} = V_i - i_e R_E$$

so R_E reduce input of T_x
& hence acts as -ve feedback

gain of amplifier reduces

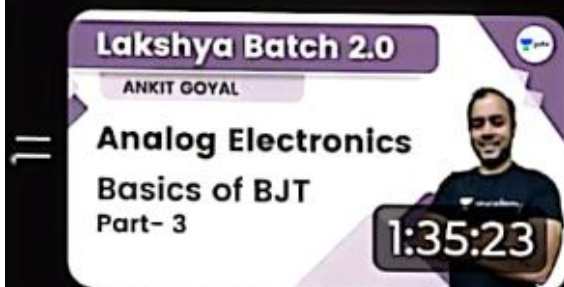
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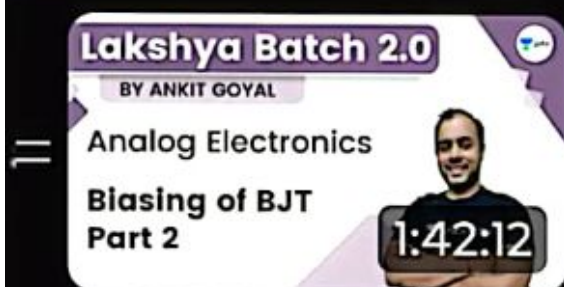
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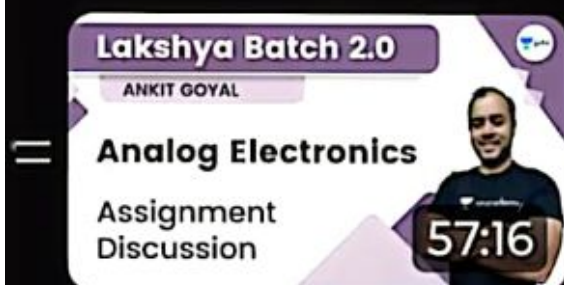
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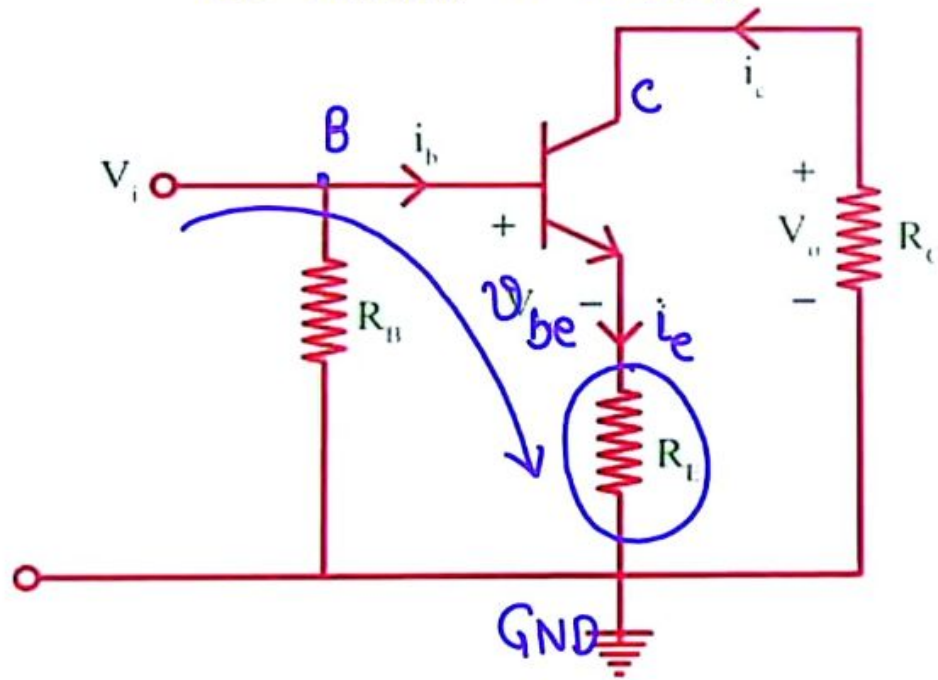


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Drawback of Emitter Biasing

ac model : $C \rightarrow \text{short}$



$$V_{be} = V_i - i_e R_E$$

so R_E reduce in
& hence acts as -
gain of amplifier



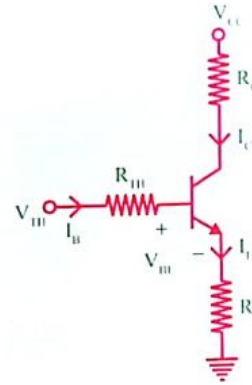
For Thevenin V_{OC} and R_{TH}



$$V_{th} = V_{cc} \frac{R_2}{R_1 + R_2}$$

or R_{TH} , ground V_{cc}

$$R_{TH} = R_1 \parallel R_2$$



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BY ANKIT GOYAL

Analog Electronics

Biasing of BJT
Part 2



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Analog Electronics

Assignment
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Analog Electronics

Compensation and
Problems on BJT
Biasing

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Analog Electronics

Current Mirror and
Voltage Regulator Circuit

Ankit Goyal

1:50:37



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LAKSHYA 2.0

Analog Electronics

AC Models of BJT
Part - 1

Ankit Goyal

1:57:34



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LAKSHYA 2.0

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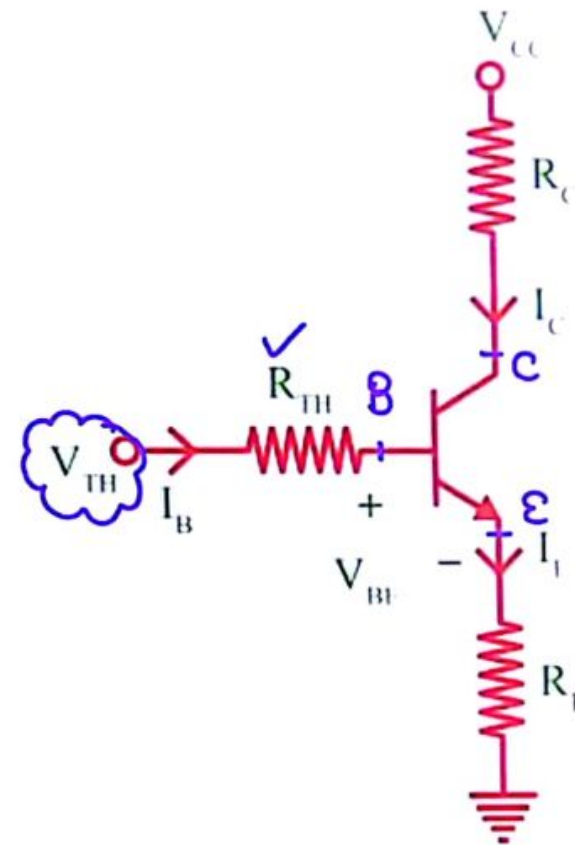


For Thevenin V_{OC} and R_{TH}

$$V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2}$$

for R_{TH} , ground V_{CC}

$$R_{TH} = R_1 \parallel R_2$$



① Apply KVL at input loop

$$V_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0$$

assume active region

$$I_E = (1 + \beta) I_B$$

& determine I_B

② $I_C = \beta I_B$; $I_E = (1 + \beta) I_B$

③ KVL in o/p loop

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$
$$V_{CE} = ?$$

$$V_{CE} > 0.2V \text{ (active)}$$

① Apply KVL at input loop

$$V_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$$

assume active region

$$I_E = (1 + \beta) I_B$$

& determine I_B

② $I_C = \beta I_B$; $I_E = (1 + \beta) I_B$

③ KVL in o/p loop

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = ?$$

$V_{CE} > 0.2V$ (active)

if $V_{CE} < 0.2$ (sat)

$$\rightarrow V_{CE} = 0.2$$

$\rightarrow$ compute I_C using last eqn

Stability factor

① Apply KVL at input loop

$$V_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$$

active region

$$\beta I_B$$

find I_B

$$I_B ; I_E = (1 + \beta) I_B$$

o/p loop

$$V_{CE} - I_E R_E = 0$$

?

$$V_{CE} > 0.2V \text{ (active)}$$

$$\text{if } V_{CE} < 0.2 \text{ (sat)}$$

$$\rightarrow V_{CE} = 0.2$$

$\rightarrow$ compute I_C using last eqn

Stability factor

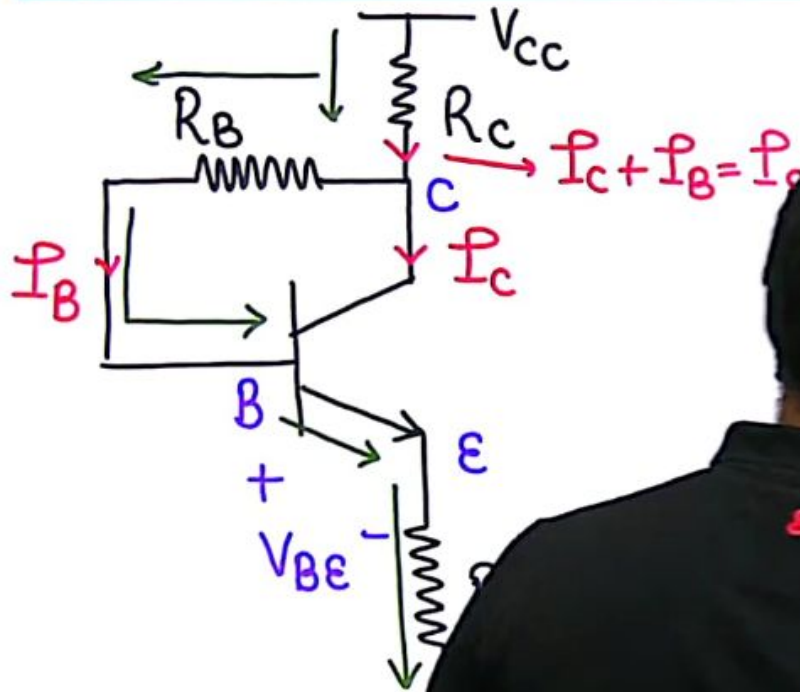
$$V_{Th} - I_B R_{Th} - V_{BE} - (I_C + I_B) R_E = 0$$

$$-\frac{\partial I_B}{\partial I_C} R_{Th} - R_E - \frac{\partial I_B}{\partial I_C} R_E = 0$$

$$\frac{\partial I_B}{\partial I_C} = \frac{-R_E}{R_{Th} + R_E}$$

$$S = \frac{1 + \beta}{1 - \beta \frac{\partial I_B}{\partial I_C}}$$
$$= \frac{1 + \beta}{1 + \beta \frac{R_E}{R_{TH} + R_E}} < 1 + \beta$$




$$V_{CC} - (I_C + I_B)R_C - I_B R_B$$

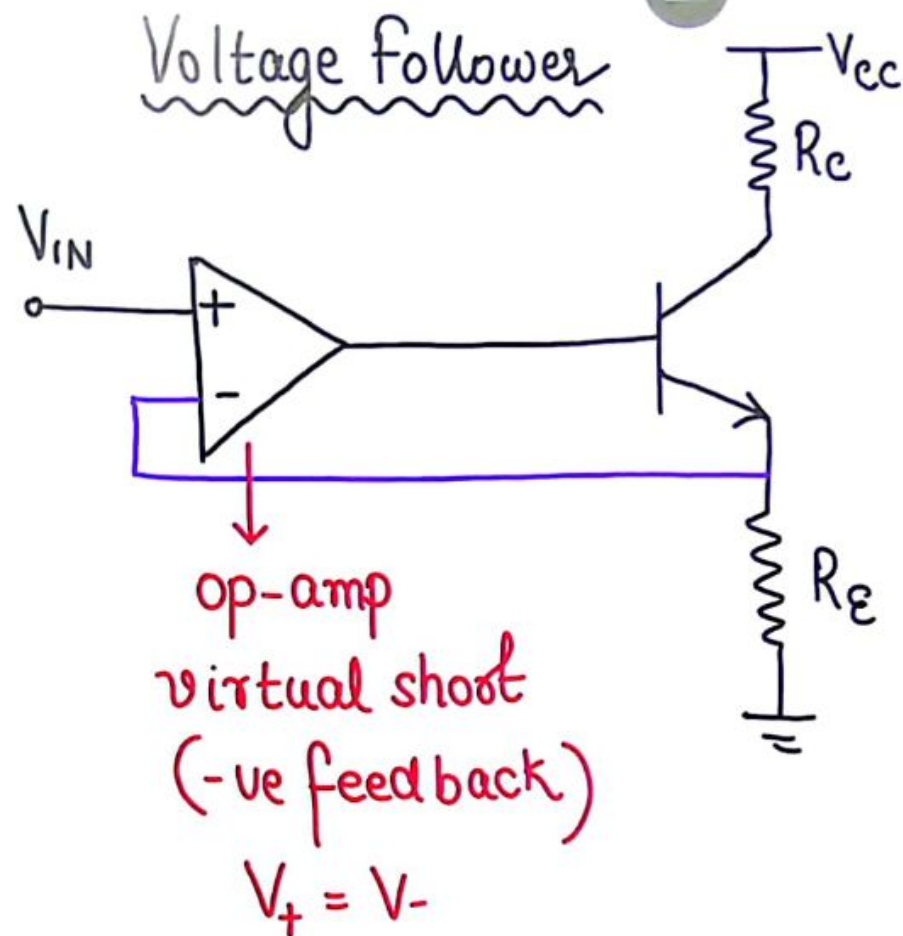
$$-V_{BE} - I_E R_E = 0$$

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} - (I_C + I_B)R_E = 0$$

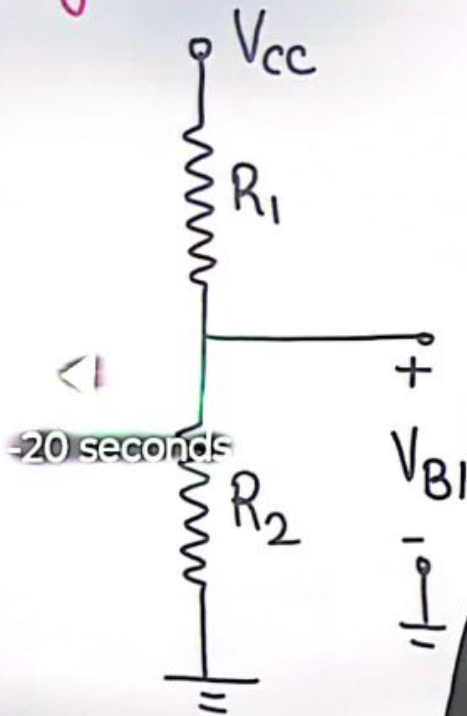
$$0 - R_c - \frac{\partial I_B}{\partial I_c} R_c - \frac{\partial I_B}{\partial I_c} R_B - 0$$

$$R_E - \frac{\partial I_B}{\partial I_C} R_E = 0$$

- to remove drift, instead of directly supplying npn transistor, we can connect it through pnp T_x to cancel V_{BE} .



Use superposⁿ
only V_{cc}



$$V_{B1} = \frac{V_{cc} R_2}{R_1 + R_2}$$

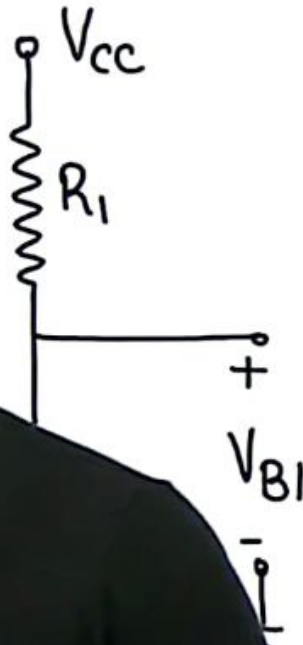
$$V_{B2} = -\frac{V_{EE} R_1}{R_1 + R_2}$$

$$V_B = V_{B1} + V_{B2}$$

$$= \frac{V_{cc} R_2}{R_1 + R_2} - \frac{V_{EE} R_1}{R_1 + R_2}$$

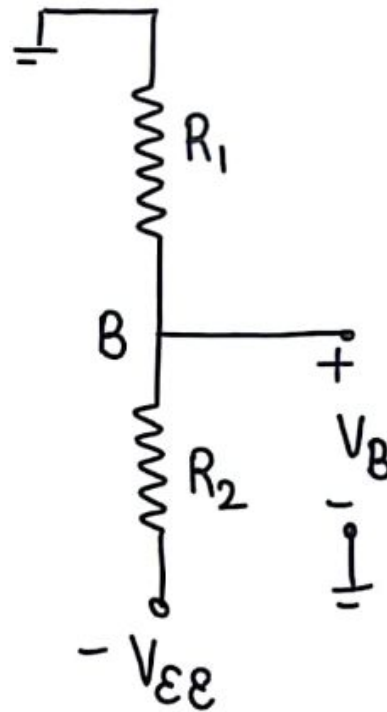
Use superpos^{tn}

only V_{CC}



$$V_{B1} = \frac{V_{CC} R_2}{R_1 + R_2}$$

only $-V_{EE}$



$$V_{B2} = -\frac{V_{EE} R_1}{R_1 + R_2}$$

$$V_B = V_{B1} + V_{B2}$$

$$= \frac{V_{CC} R_2}{R_1 + R_2} - \frac{V_{EE} R_1}{R_1 + R_2}$$

$$R_{TH} = R_1 \parallel R_2$$

$$4.18 - 7.67 I_B - 0.7 - 1 I_E = 0$$

$$3.48 = 7.67 I_B + 51 I_B$$

$$I_B = 0.0594 \text{ mA}$$

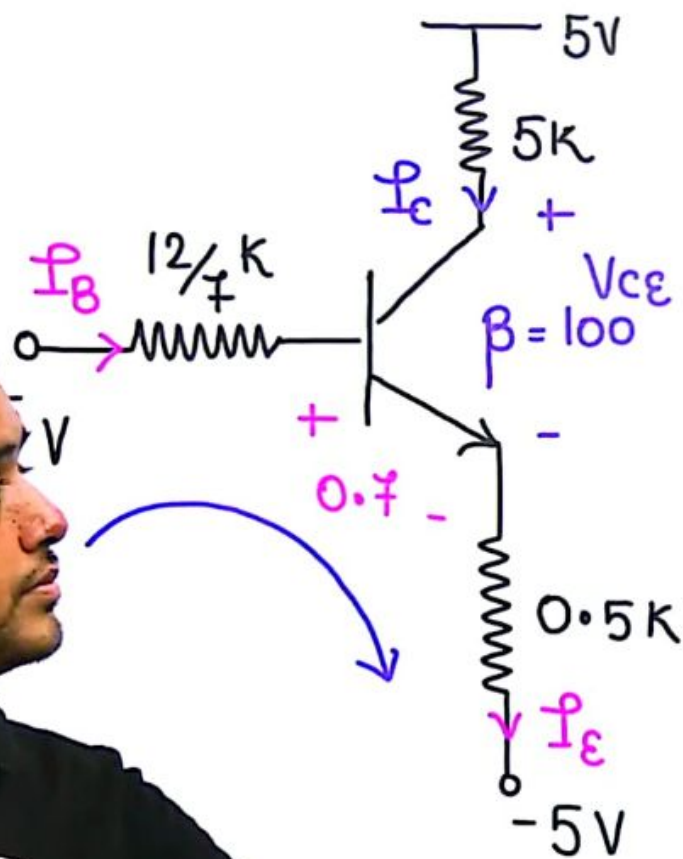
$$I_C = \beta I_B = 2.96 \text{ mA}$$

$$18 - 2.2 I_C - V_{CE} - 1 I_E = 0$$

$$V_{CE} = 18 - 2.2 \times 2.96 - 51 \times 0.0594$$

$$= 8.468 \text{ V} > 0.2$$

(active)



$$-25/7 - I_B \times 12/7 - 0.7 - I_E \times 0.5 = -5$$

$$-25/7 + 5 - 0.7 = I_B \left[\frac{12}{7} + 101 \times 0.5 \right]$$

$$I_B = 0.139 \mu A$$

$$I_C = \beta I_B = 1.39 \text{ mA}$$

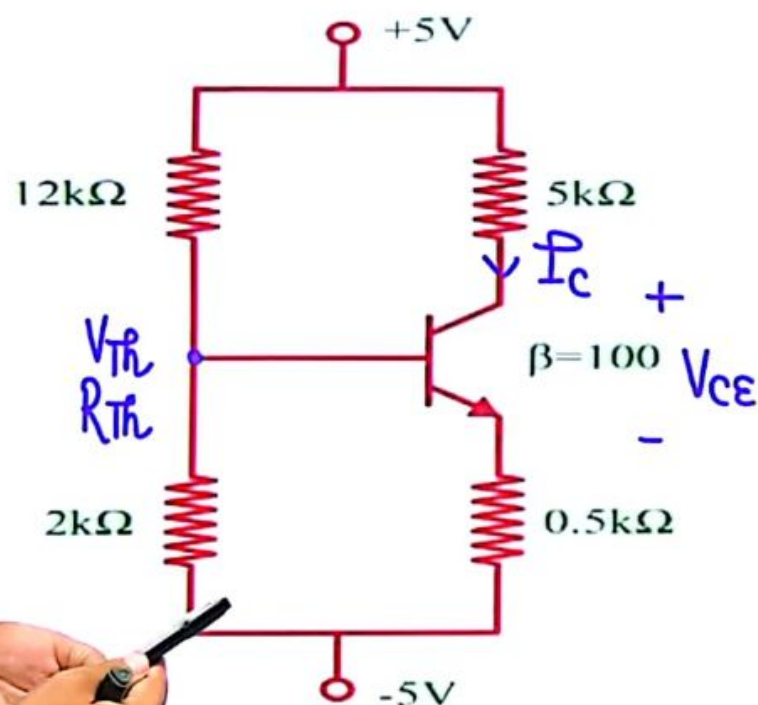
$$5 - 5 I_C - V_{CE} - 0.5 I_E = -5$$

$$V_{CE} = 10 - 5 \times 1.39 - 0.5 \times 101 \times 0.139 =$$



Question-07

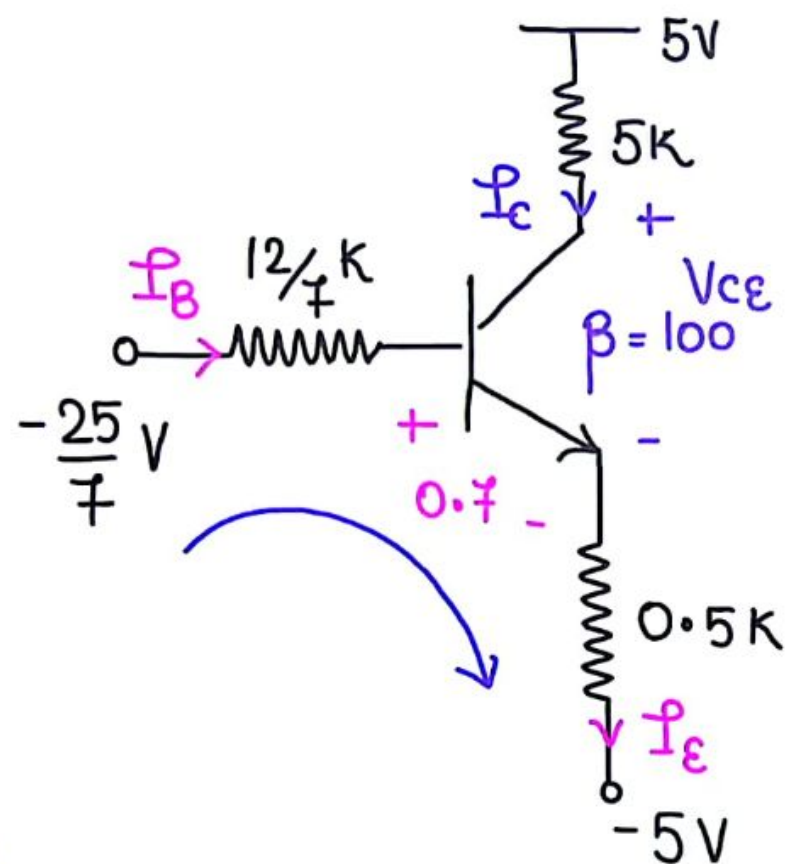
The current gain of the transistor shown in the circuit below is $\beta = 100$. The values of Q-point (I_{CQ} , V_{CEQ}) is



$$V_{TH} = 5 \times \frac{2}{12+2} - 5 \times \frac{12}{12+2} = -\frac{50}{14} \\ = -\frac{25}{7} \text{ volt}$$

$$R_{TH} = 12 \parallel 2 = \frac{12}{7} \text{ k}\Omega$$

- (a) (1.8mA, 2.1V)
- (b) (1.4mA, 2.3V)
- (c) (1.4mA, 1.8V)
- (d) (1.8mA, 1.4mV)



$$-\frac{25}{7} - I_B \times \frac{12}{7} - 0.7 - I_E \times 0.5 = -5$$

$$-\frac{25}{7} + 5 - 0.7 = I_B \left[\frac{12}{7} + 101 \times 0.5 \right]$$

$$I_B = 0.0139 \text{ mA}$$

$$I_C = \beta I_B = 1.39 \text{ mA}$$

$$5 - 5 I_C - V_{CE} - 0.5 I_E = -5$$

$$V_{CE} = 10 - 5 \times 1.39 - 0.5 \times 101 \times 0.0139 = 2.34 \text{ V}$$

What is Current Mirror



$$I_{copy} = I_{ref}$$

$$= N \times I_{ref}$$

$$= \frac{I_{ref}}{N}$$

So, now let's see, how it can be designed using the BJT.

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BJT- Current Mirror Explained

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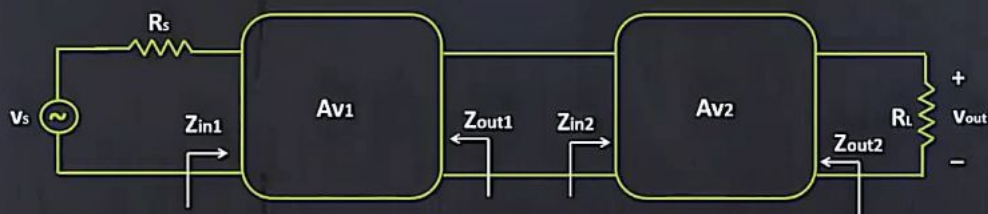


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no need for the intro music

BJT- Multistage Amplifier



15:38



BJT- Multistage Amplifier Explained (with Example)



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BJT Current Mirror



Because the collector current I_C is the function of the voltage V_{be} .

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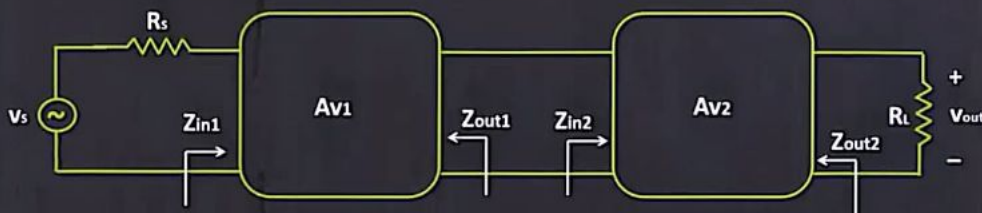
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BJT- Multistage Amplifier



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BJT- Multistage Amplifier Explained (with Example)

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$$I_C \approx I_S \exp\left(\frac{V_{BE}}{V_T}\right)$$

That means by controlling this voltage V_{be} , we can control this collector current.

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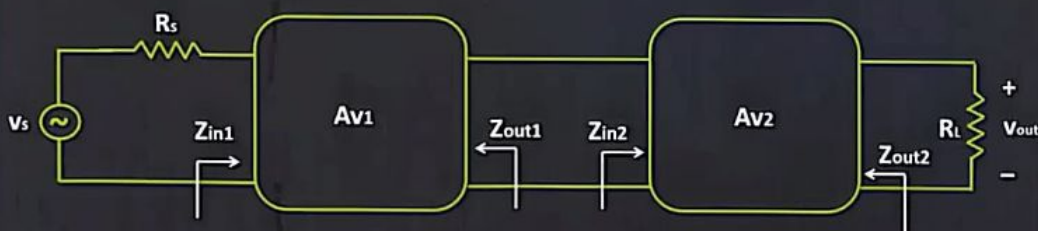
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Very confusing. First of all, What is V_t at 3:00 ?

BJT- Multistage Amplifier



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BJT- Multistage Amplifier Explained (with Example)

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BJT Current Mirror



So, here as you can see, the collector and the base terminal of the transistor are connected

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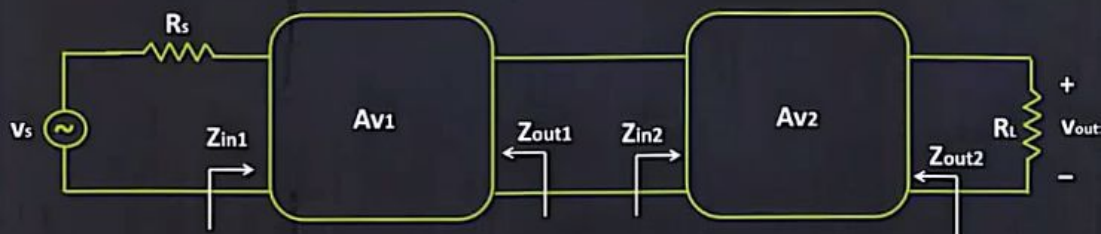
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BJT Current Mirror



$$I_{C1} \approx I_{S1} \exp\left(\frac{V_{BE}}{V_T}\right)$$

$$I_{REF} \approx I_{S(REF)} \exp\left(\frac{V_{BE}}{V_T}\right)$$

-So, here this I_{ref} can be given as $I_S (ref)^*$

$$V_{BE(REF)} = V_{BE1} \exp(V_{BE} / V_T)$$

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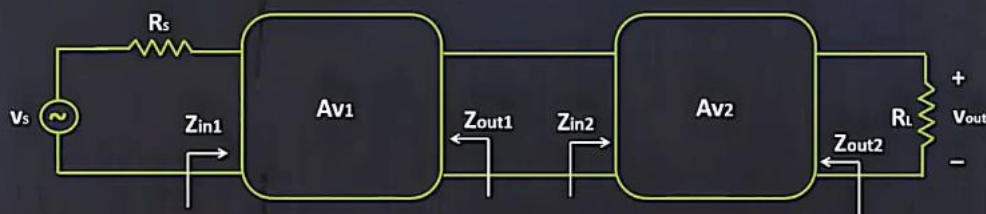


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My professor took 3 lecture(around 3 hr) to explain current mirror circuit but u explained same thing in ju...

BJT- Multistage Amplifier



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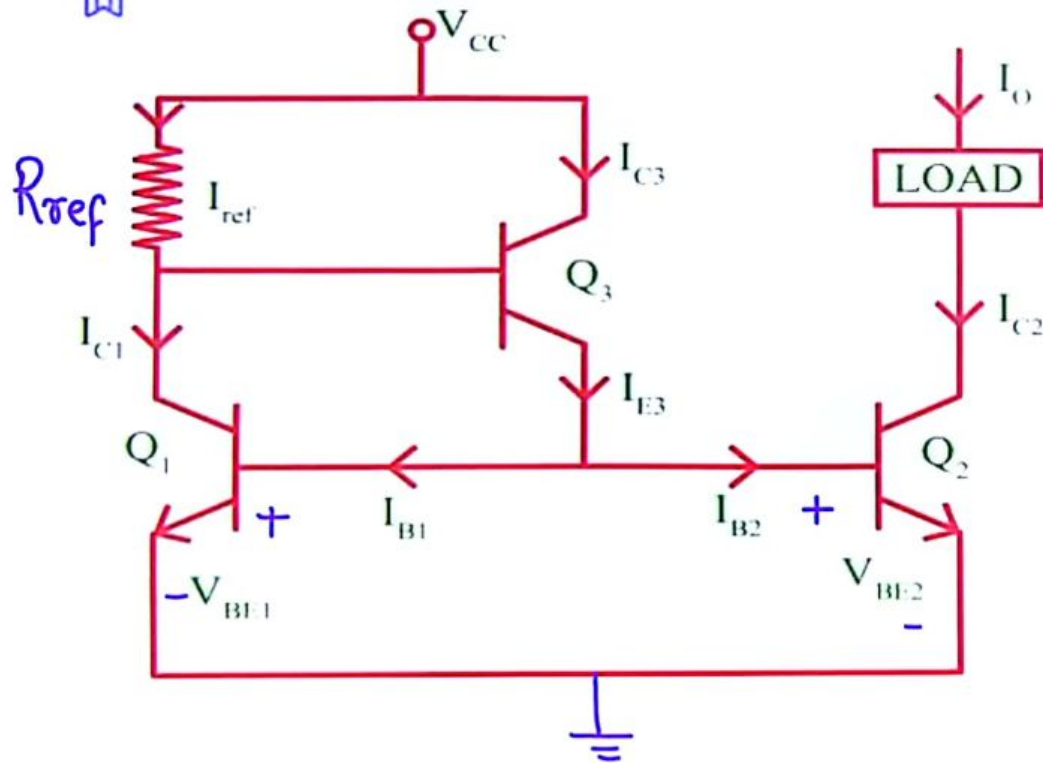
BJT- Multistage Amplifier Explained (with Example)



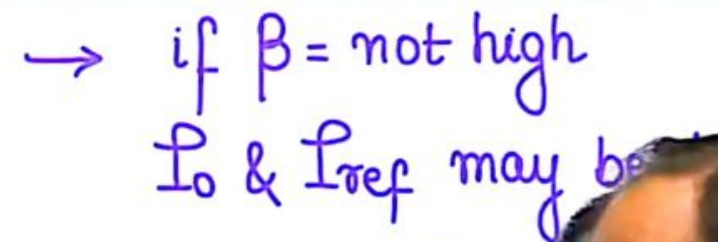
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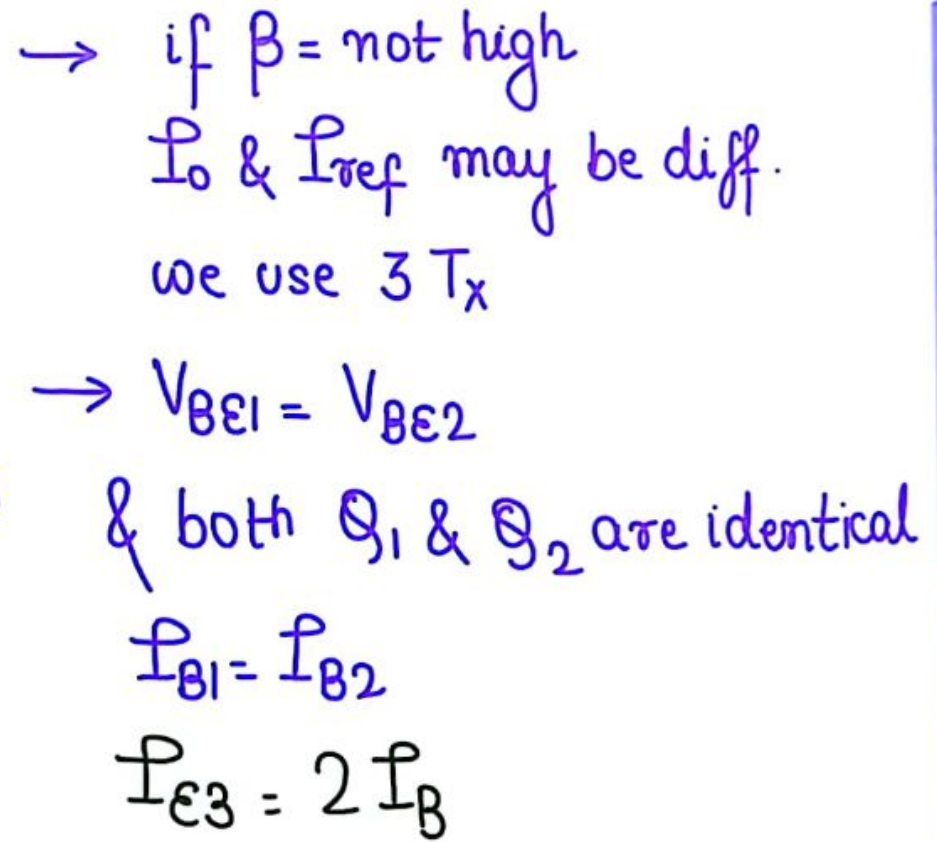
→ if β = not high
 I_o & I_{ref} diff.
we use
→ $V_{BE1} = V_{BE2}$
& both Q1 & Q2 identical
 $I_{B1} = I_{B2}$


$$\rightarrow V_{BE1} = V_{BE2}$$

& both Q_1 & Q_2 are

$$I_{B1} = I_{B2}$$

$$I_{E3} = 2 I_B$$



$$I_{\text{ref}} = I_{B3} + I_{C1}$$

$$= I_{B3} + \beta I_B$$

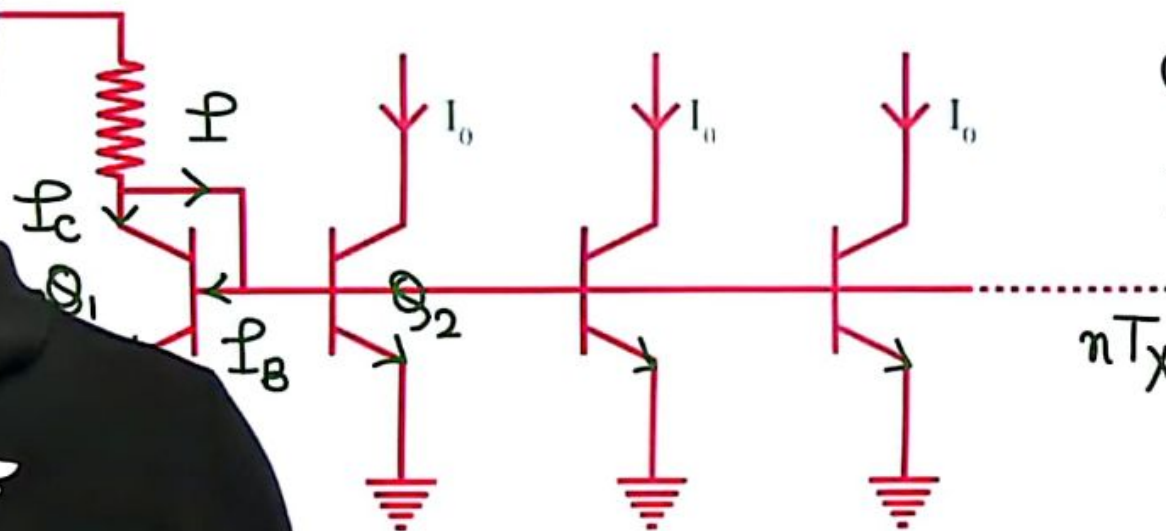
$$= I_{B3} + \beta I_B$$

$$\begin{aligned} I_0 = I_{C2} &= \beta I_B \\ &= \frac{\beta I_{\text{ref}}}{\beta + \frac{2}{1 + \beta_3}} \end{aligned}$$

$$I_0 = \frac{I_{\text{ref}}}{1 + \frac{2}{\beta(1 + \beta_3)}}$$



Multiple Current Source



$$I = n I_B$$

all V_{BE} are equal &
all T_x are identical

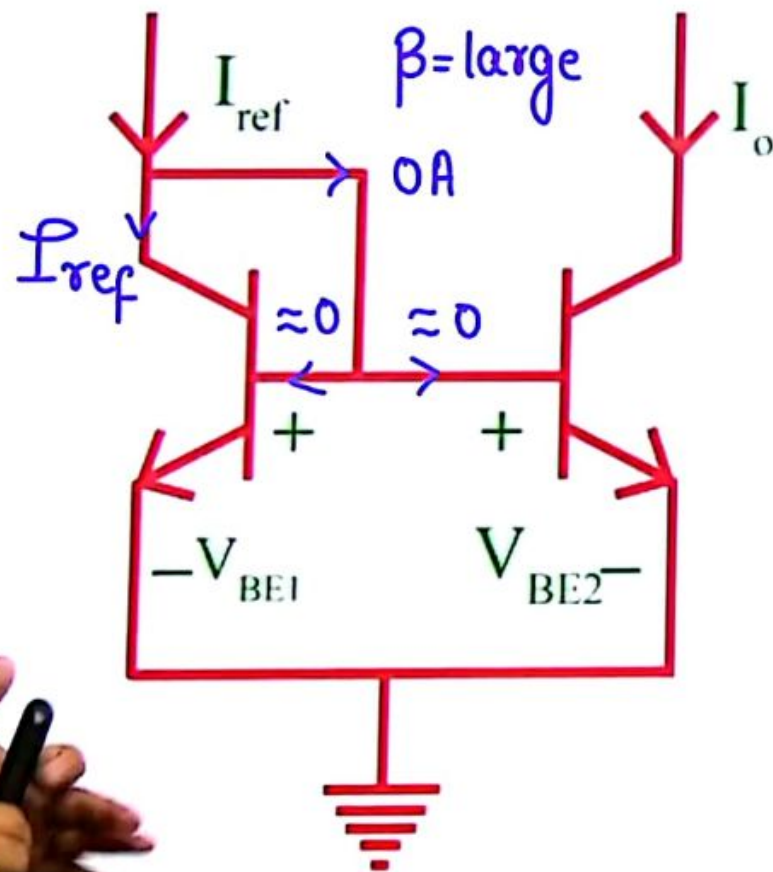
$$\begin{aligned} I_{ref} &= I_C + I \\ &= \beta I_B + n I_B \end{aligned}$$

$$I_B = \frac{I_{ref}}{n + \beta}$$

$$I_0 = \beta I_B = \frac{I_{ref}}{1 + n/\beta}$$



Trans-linear property



$$I_{E1} = I_{E01} \left[e^{V_{BE1}/V_T} - 1 \right]$$

$$I_{E2} = I_{E02} \left[e^{V_{BE2}/V_T} - 1 \right]$$

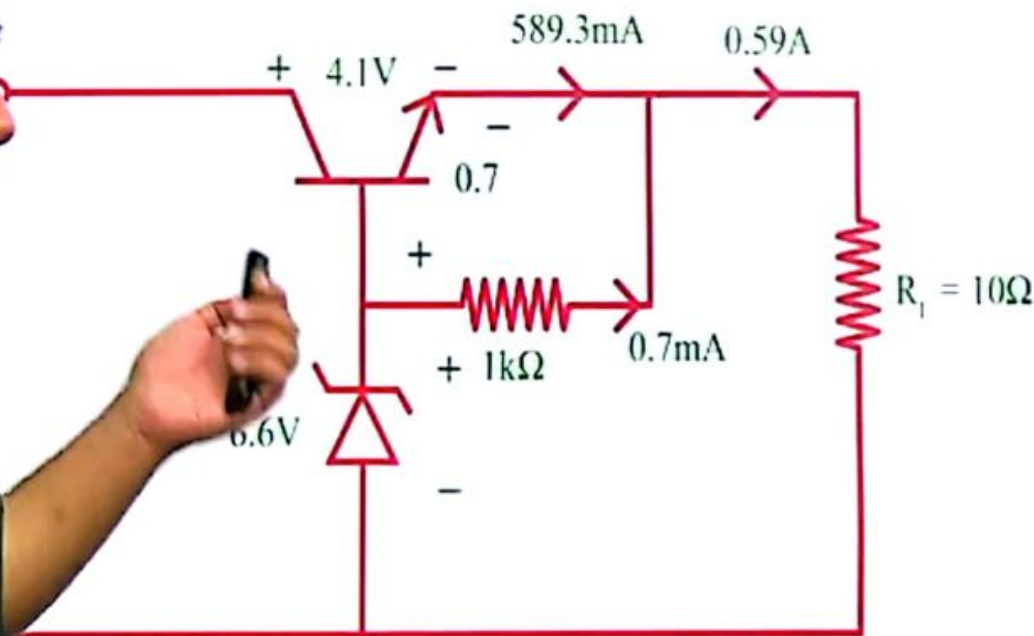
$$\because V_{BE2} = V_{BE1}$$

$$\frac{I_{E1}}{I_{E2}} = \frac{I_{E01}}{I_{E02}} = \frac{A_1}{A_2}$$

T_x are identical in all respects except CSA

$$\beta = \text{large}; \quad I_o / I_{ref} = A_2 / A_1$$

The three terminal voltage regulator is connected to a 10Ω resistance, as shown in the figure. If input voltage is $10V$. What is the power dissipation in the transistor

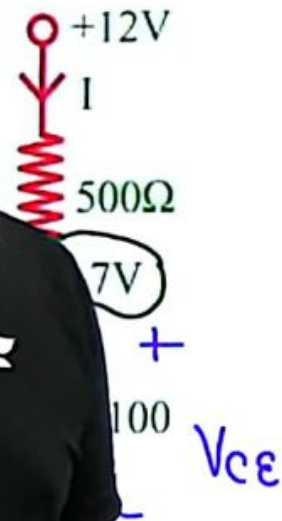




Student's Assignment

Question-01

Find the operating point of transistor, $\beta=100$, $V_Z = 5V$



$$I = \frac{12 - 5.7}{500} = 12.6 \text{ mA}$$

$$I_E = 12.6 \text{ mA} = \frac{(\beta + 1)}{\beta} I_C$$

$$I_C = 12.475 \text{ mA}$$

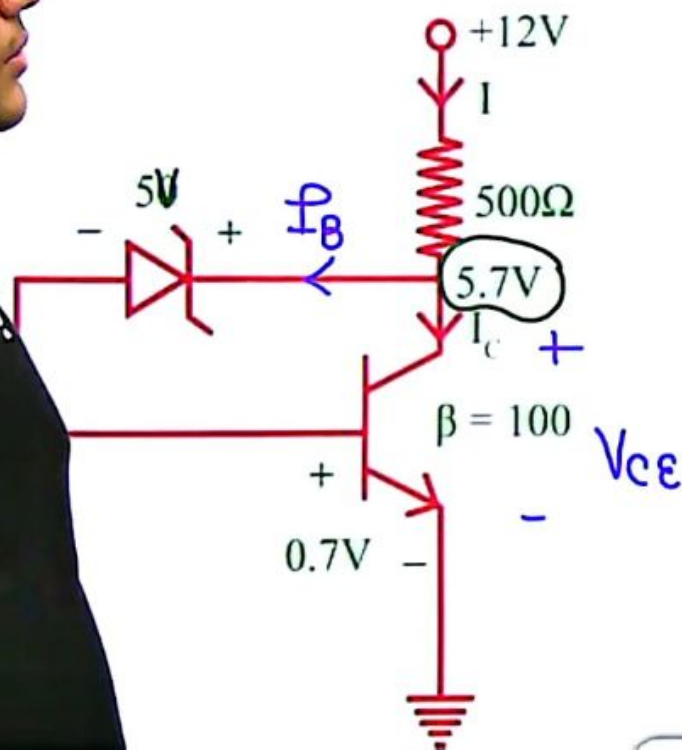
$$V_{CE} = 5.7 \text{ V}$$



Student's Assignment

Question-01

Find the Q point of transistor, $\beta = 100$, $V_Z = 5V$



$$I = \frac{12 - 5.7}{500} = 12.6 \text{ mA}$$

$$I_E = 12.6 \text{ mA} = \frac{(\beta + 1)}{\beta} I_C$$

$$I_C = 12.475 \text{ mA}$$

$$V_{CE} = 5.7 \text{ V}$$

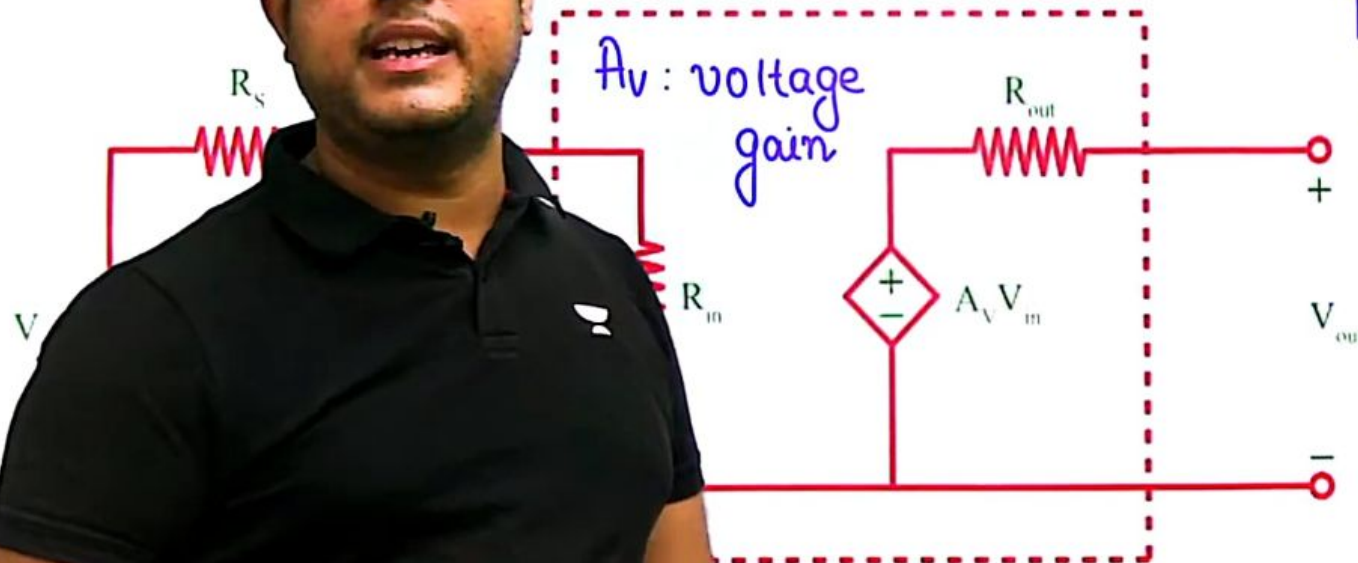


BJT AMPLIFIERS

11:41

Voltage amplifier

• voltage controlled voltage source.



• increase the amplitude & power of input ac signal.

power of a sinusoidal signal

$$A \sin \omega t$$

$$P = \frac{A^2}{2}$$

Amplifier Parameters

- Input resistance

$$R_{IN} = V_{IN} / I_{IN}$$

- Output resistance

$$R_{OUT} = V_{OUT} / I_{OUT}$$

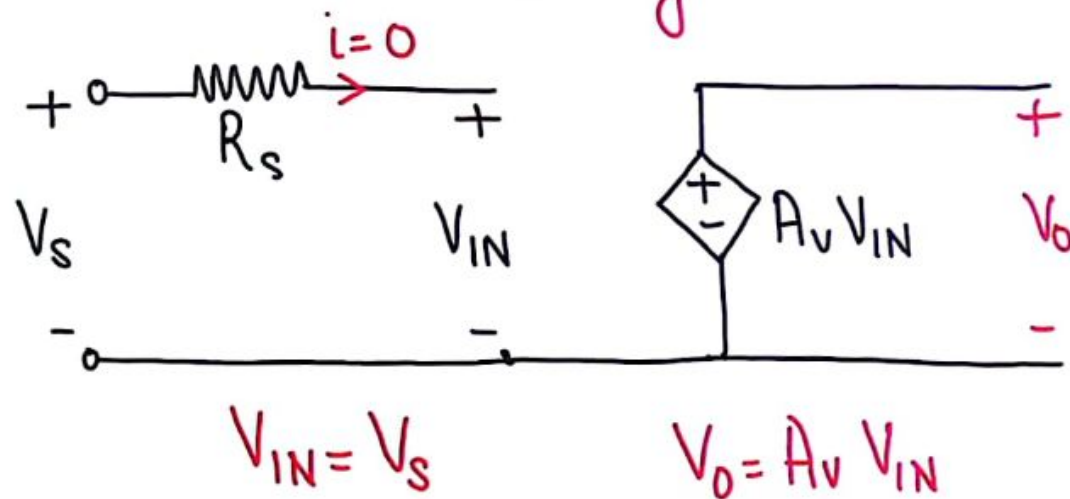
Voltage gain

$$A_V = V_{OUT} / V_{IN}$$

Ideal voltage amp^r

$$R_{IN} \rightarrow \infty \quad R_{OUT} \rightarrow 0$$

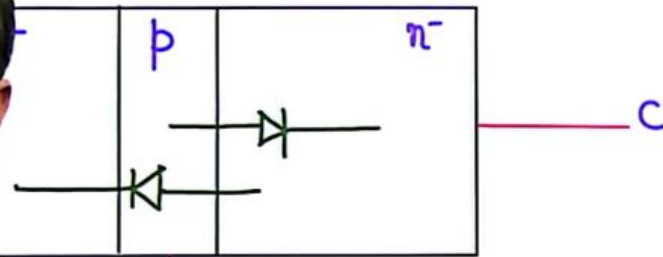
$A_V = \text{high}$



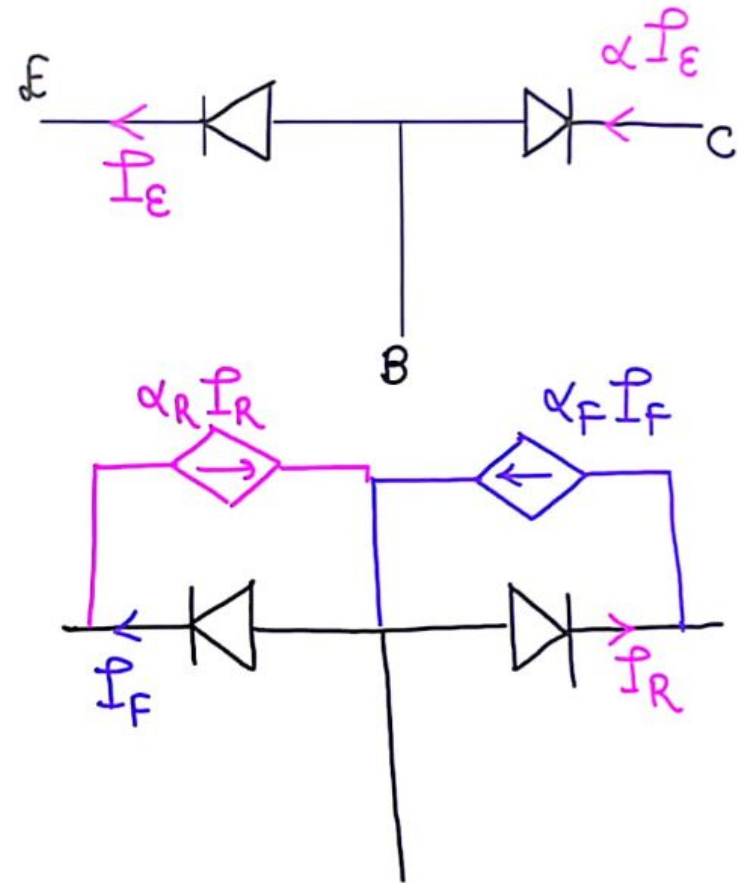


Modelling of BJT

(1) Ebers-Moll Model



α_F : forward current gain
reverse current gain



$$\begin{aligned} I_E &= I_F - \alpha_R I_R \\ &= I_{ES} \left[e^{V_{BE}/V_T} - 1 \right] \\ &\quad - \alpha_R I_{CO} \left[e^{V_{BC}/V_T} - 1 \right] \end{aligned}$$

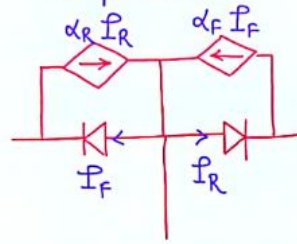
$$\begin{aligned} I_C &= \alpha_F I_F - I_R \\ &= \alpha_F I_{ES} \left[e^{V_{BE}/V_T} - 1 \right] \\ &\quad - I_{CO} \left[e^{V_{BC}/V_T} - 1 \right] \end{aligned}$$



Saturation

EB: forward

CB: forward

Cut-off

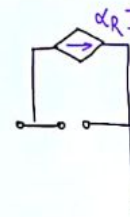
EB: reverse

CB: reverse

Inverse active

EB: reverse

CB: forward

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Saturation
EB: forward
CB: forward
 $\alpha_F I_F$

Cut-off
EB: reverse
CB: reverse
 $I_F = 0$ $I_R = 0$

Inverse active
EB: reverse
CB: forward
 $\alpha_R I_R$

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Part - 2

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Analog Electronics

AC Models of BJT
Part - 3

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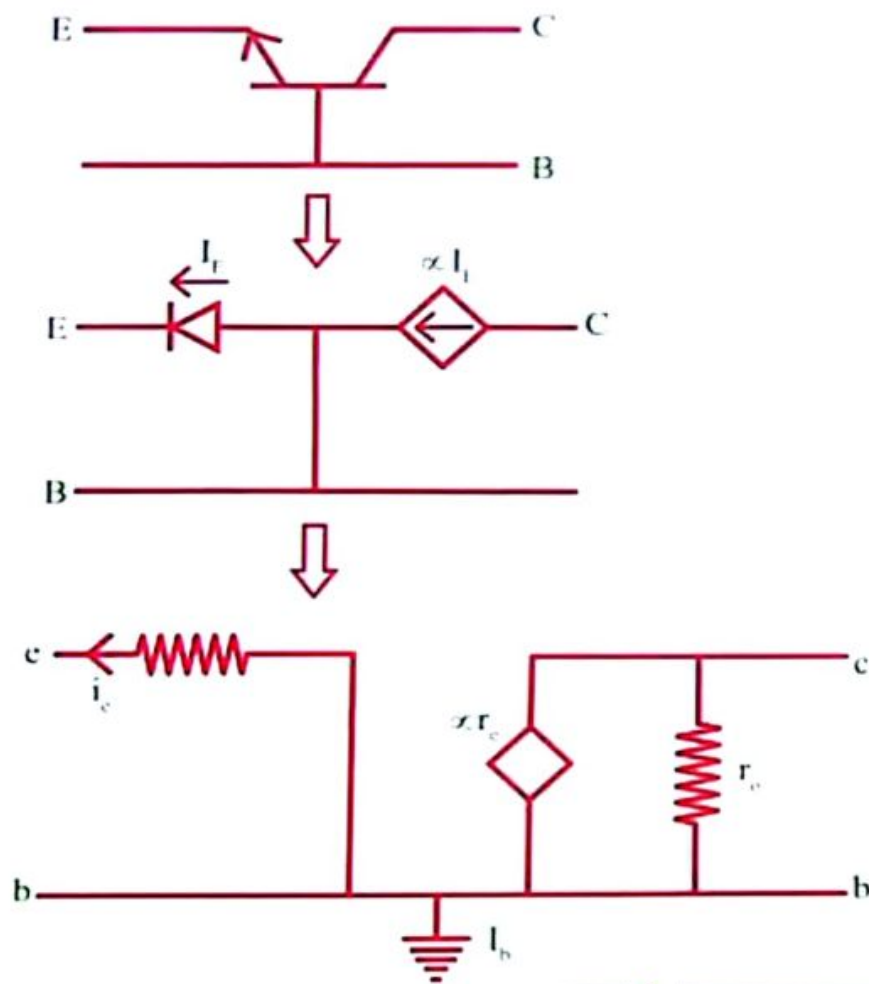
Analog Electronics

AC Models of BJT
Part - 4

1:36:55

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Electronics | AC Models of B...
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Common Base Configuration



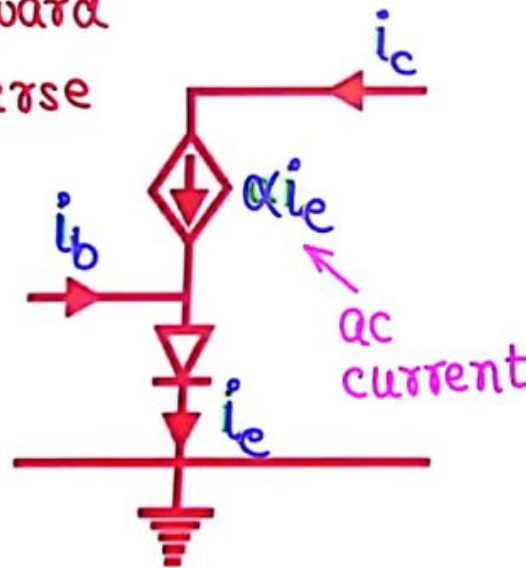


(2) R_E Transistor model:

- In active mode of operation if the diode is replaced by a resistance, then R_E model is obtained

Common Emitter configuration

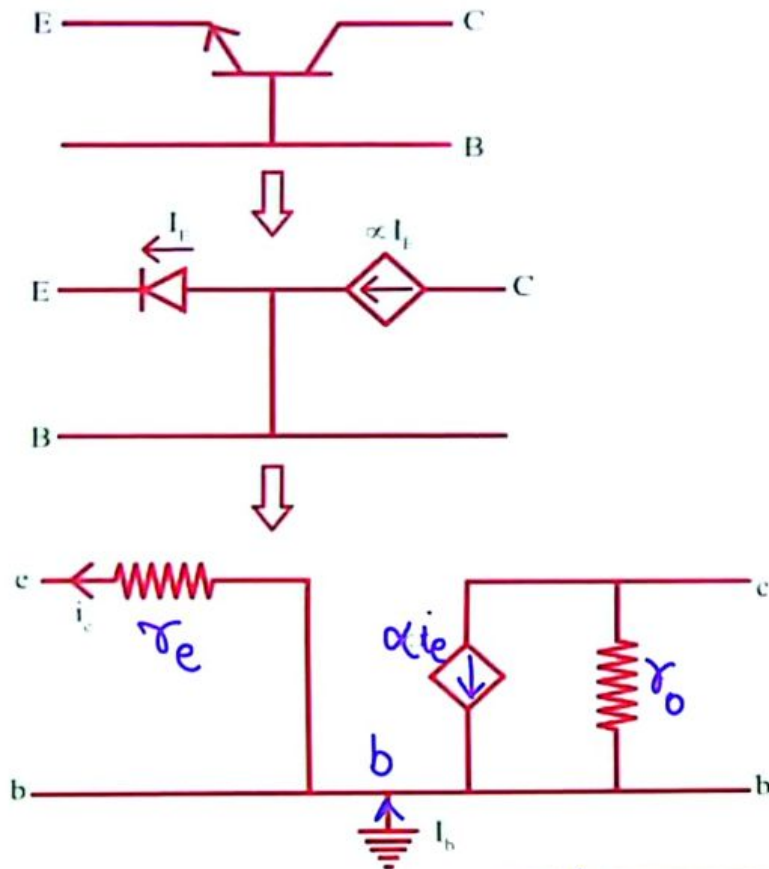
active: $E B \rightarrow$ forward
 $C B \rightarrow$ reverse



when a diode is forward biased then it can be replaced by a resistance called Diffusion Resistance (small signal)



Common Base Configuration

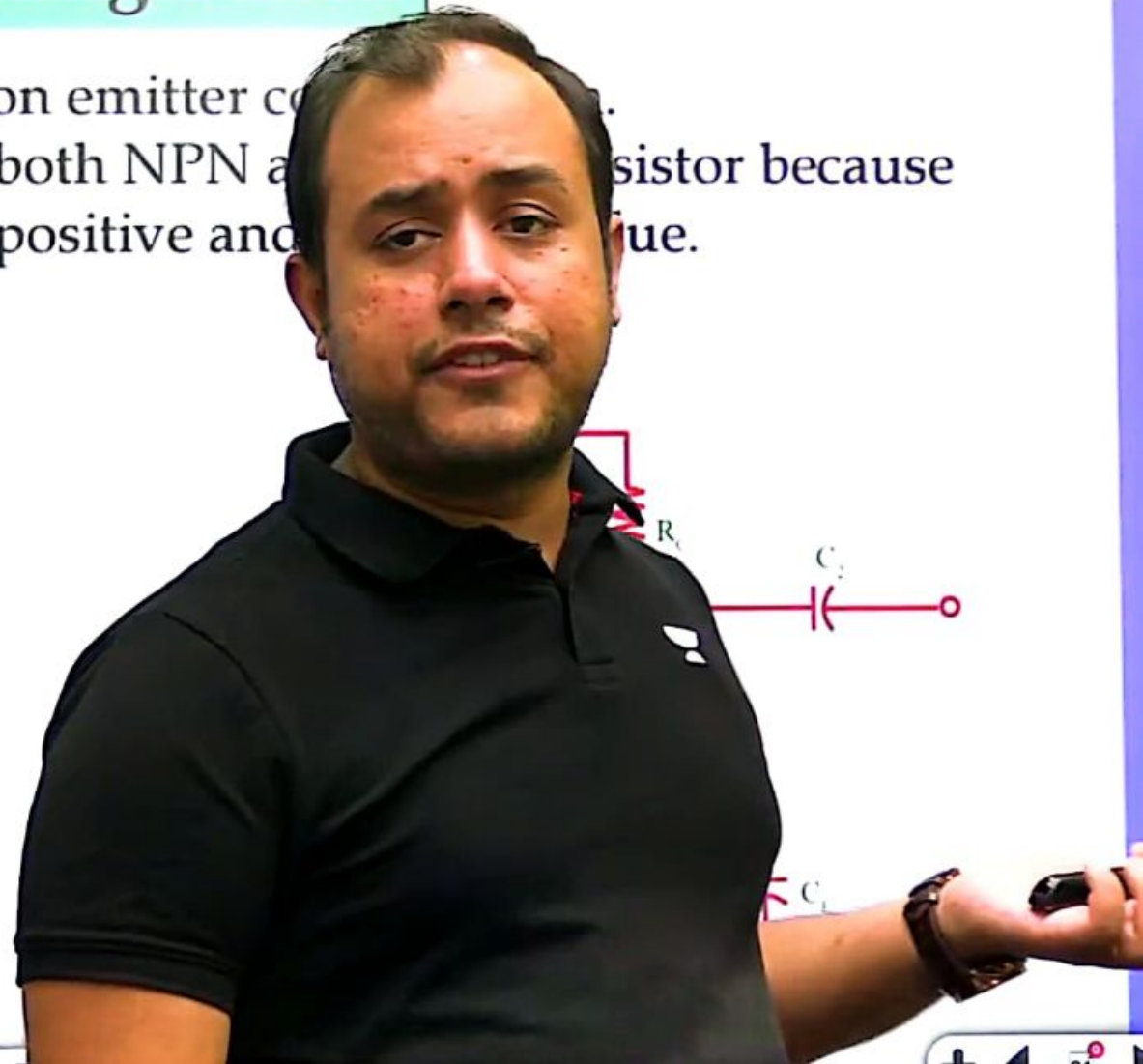




Common Collector Configuration

- The model is same as common emitter configuration.
- The AC analysis is same for both NPN and PNP transistor because input signal varies between positive and negative value.

Voltage Divider Bias





Common Collector Configuration

- The model is same as common emitter configuration
- The AC analysis is same for both NPN and PNP for because input signal varies between positive and negative

Voltage Divider Bias

dc analysis: $C \rightarrow \text{open}$

$$V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2}$$

$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2}$$

KVL in BE loop; $I_B = ?$

V_{in}

al



Common Collector Configuration

- The model is same as common emitter configuration.
- The AC analysis is same for both NPN and PNP transistor because output signal varies between positive and negative value.

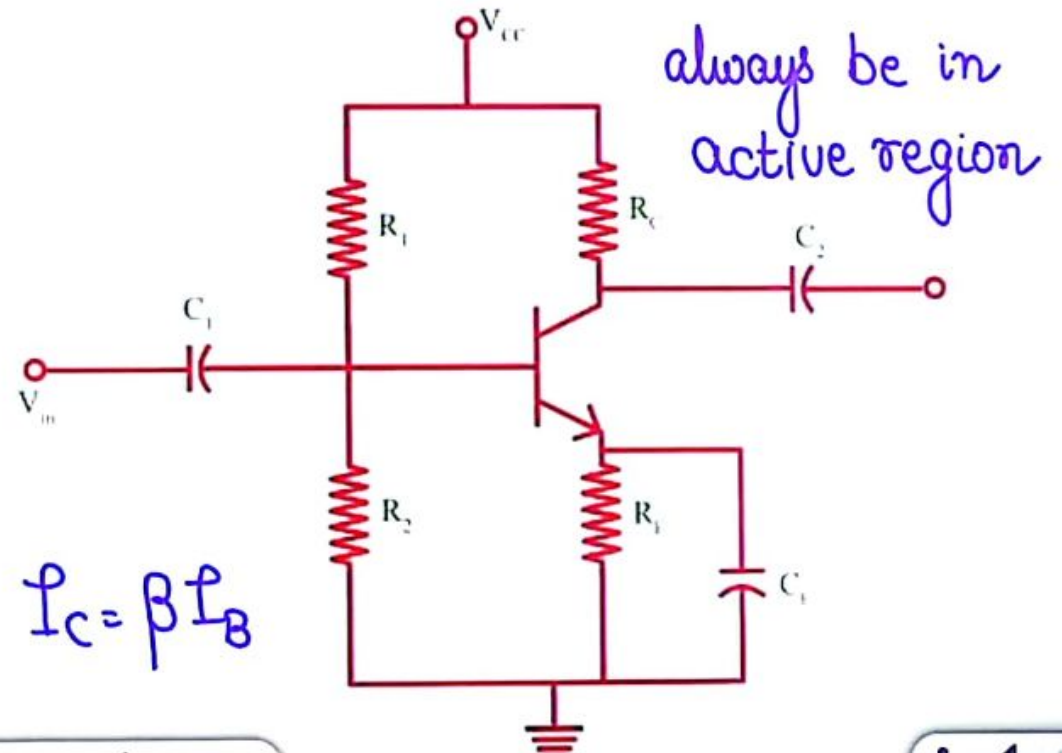
Voltage Divider Bias

dc analysis: $C \rightarrow \text{open}$

$$V_{th} = V_{cc} \frac{R_2}{R_1 + R_2}$$

$$R_{th} = \frac{R_1 R_2}{R_1 + R_2}$$

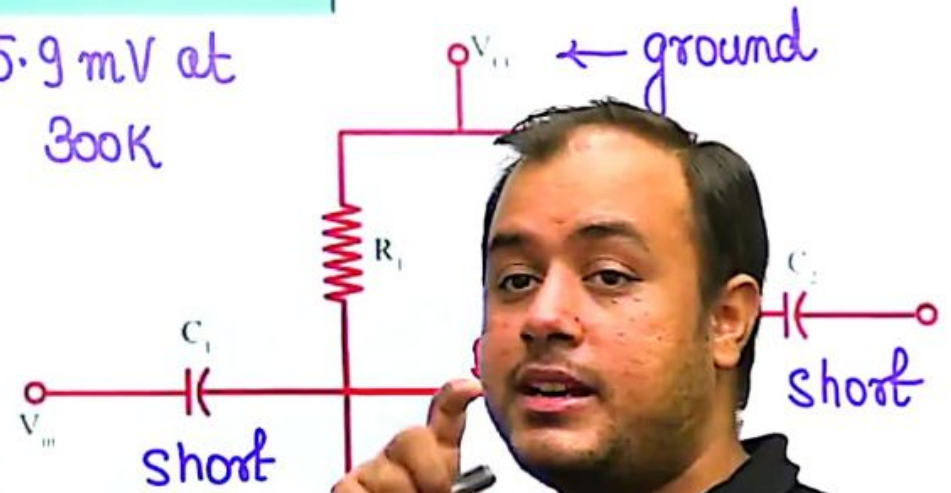
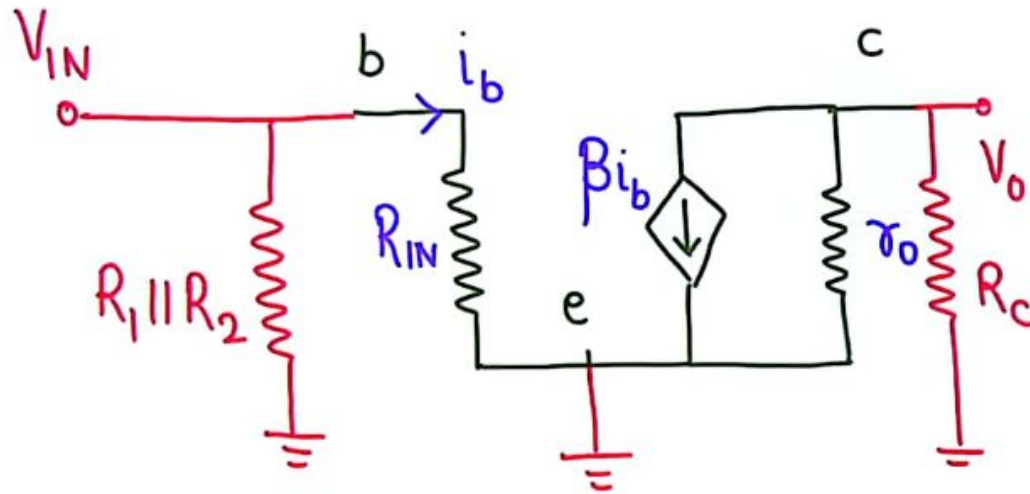
In BE loop; $I_B = ?$ $I_C = \beta I_B$





Voltage Divider Bias

- $R_{IN} = (1 + \beta) r_e = V_T / I_B$; $V_T = 25.9 \text{ mV}$ at 300K
- $r_o = V_A / I_C$





Output impedance (Z_o)

$$Z_{OUT} = V_{OUT} / I_{OUT} \Big|_{V_{IN} = 0}$$

$$\text{if } V_{IN} = 0 \quad i_b = 0$$

$$\beta i_b = 0$$

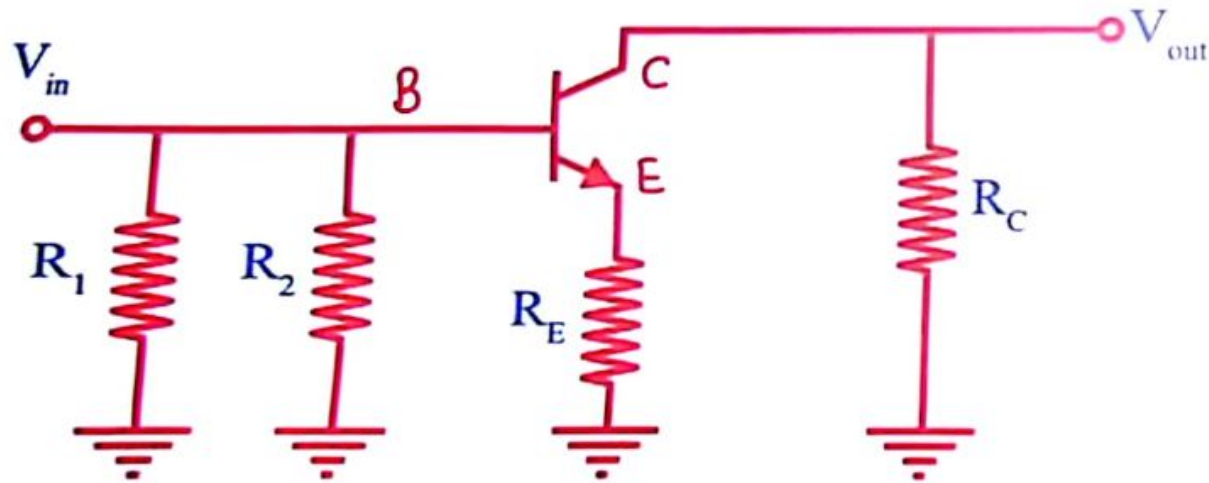
$$Z_{OUT} = r_o \parallel R_C \approx R_C$$

r_o is very high

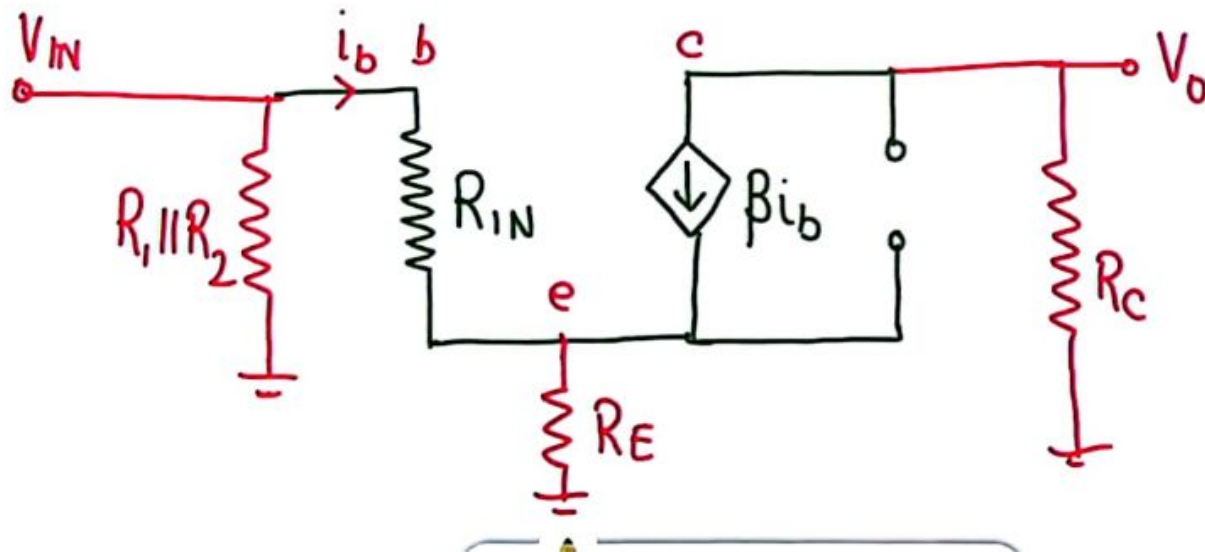




With-out By-Pass Capacitor



assume $r_o = \text{high}$
ignore r_o





Voltage Gain A_V

$$V_o = -\beta i_b R_c$$

$$V_{IN} - i_b R_{IN} - (1+\beta) i_b R_E = 0$$

$$i_b = \frac{V_{IN}}{R_{IN} + (1+\beta) R_E}$$

$$V_o = -\frac{\beta V_{IN} R_c}{R_{IN} + (1+\beta) R_E}$$

$$\frac{V_o}{V_{IN}} = \frac{-\beta R_c}{(1+\beta) r_e + (1+\beta) R_E}$$

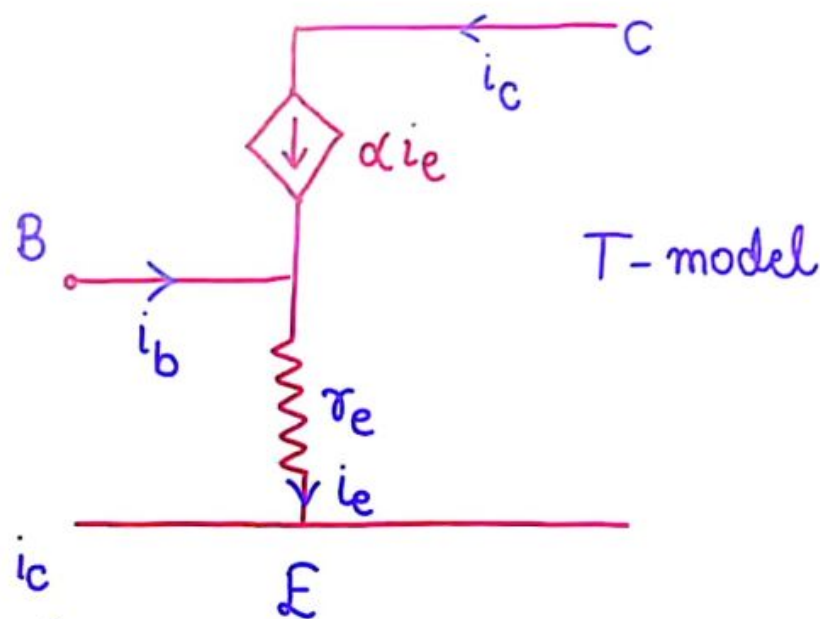
$\beta = \text{large}$

$$A_V = \frac{-R_c}{r_e + R_E} \quad \text{gain reduces}$$

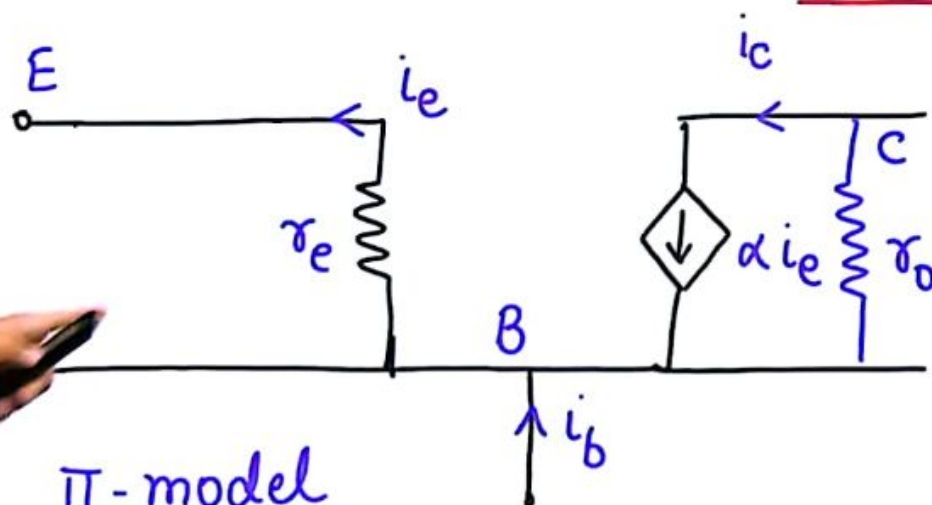
by pass cap: $-R_c / r_e$

$$r_e = V_T / I_E$$

I_E : Q-point emitter current



T-model



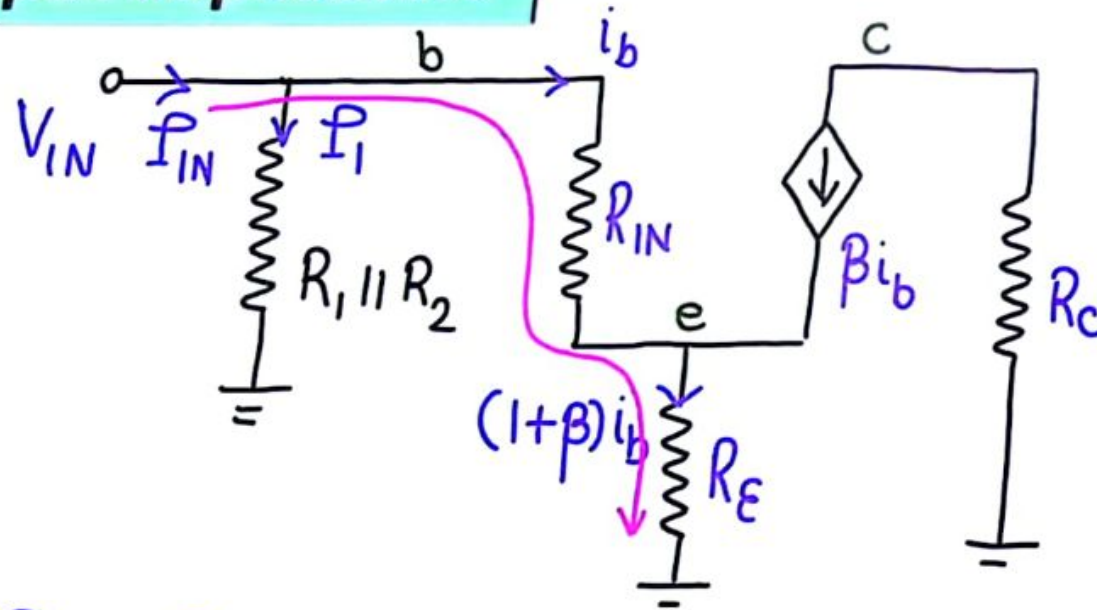
π -model

Common-base

$$r_o = V_A / I_{CQ}$$

V_A : early voltage

Input Impedance



while referring a res
from $E \rightarrow B$
multiply by $(1+\beta)$

$$I_{IN} = \frac{V_{IN}}{R_1 \parallel R_2} + \frac{V_{IN}}{R_{IN} + (1+\beta)R_E}$$

$$I_{IN} = I_1 + i_b$$

$$I_1 = \frac{V_{IN}}{R_1 \parallel R_2}$$

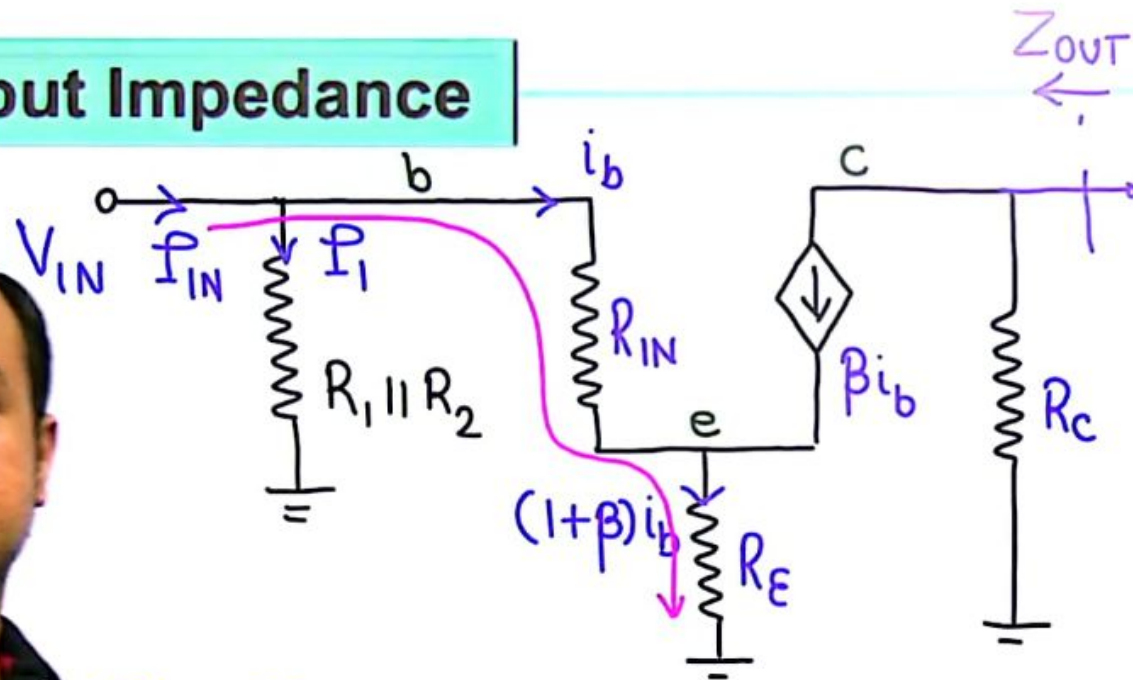
$$V_{IN} - i_b R_{IN} - (1+\beta)i_b R_E = 0$$

$$i_b = \frac{V_{IN}}{R_{IN} + (1+\beta)R_E}$$

$$\frac{V_{IN}}{I_{IN}} = R_1 \parallel R_2 \parallel [R_{IN} + (1+\beta)R_E]$$



Input Impedance



while referring a res
from $E \rightarrow B$
multiply by $(1+\beta)$

$$I_{IN} = \frac{V_{IN}}{R_1 \parallel R_2} + \frac{V_{IN}}{R_{IN} + (1+\beta)R_E}$$

$$I_{IN} = I_1 + i_b$$

$$I_1 = \frac{V_{IN}}{R_1 \parallel R_2}$$

$$V_{IN} - i_b R_{IN} - (1+\beta)i_b R_E = 0$$

$$i_b = \frac{V_{IN}}{R_{IN} + (1+\beta)R_E}$$

$$\frac{V_{IN}}{I_{IN}} = R_1 \parallel R_2 \parallel [R_{IN} + (1+\beta)R_E]$$



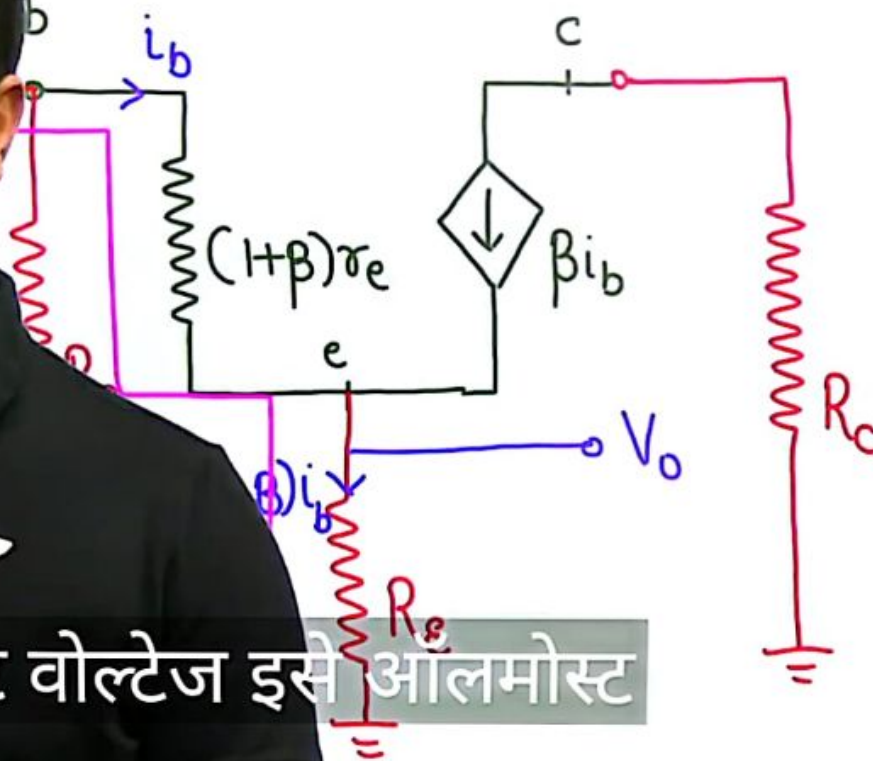
Output Impedance

$$Z_{OUT} = R_C$$





Voltage Gain A_V



$$V_{IN} - i_b(1+\beta)r_e - (1+\beta)i_b R_E = 0$$

$$i_b = \frac{V_{IN}}{(1+\beta)(r_e + R_E)}$$

$$\begin{aligned} V_O &= (1+\beta)i_b R_E \\ &= \frac{(1+\beta)R_E V_{IN}}{(1+\beta)(r_e + R_E)} \end{aligned}$$

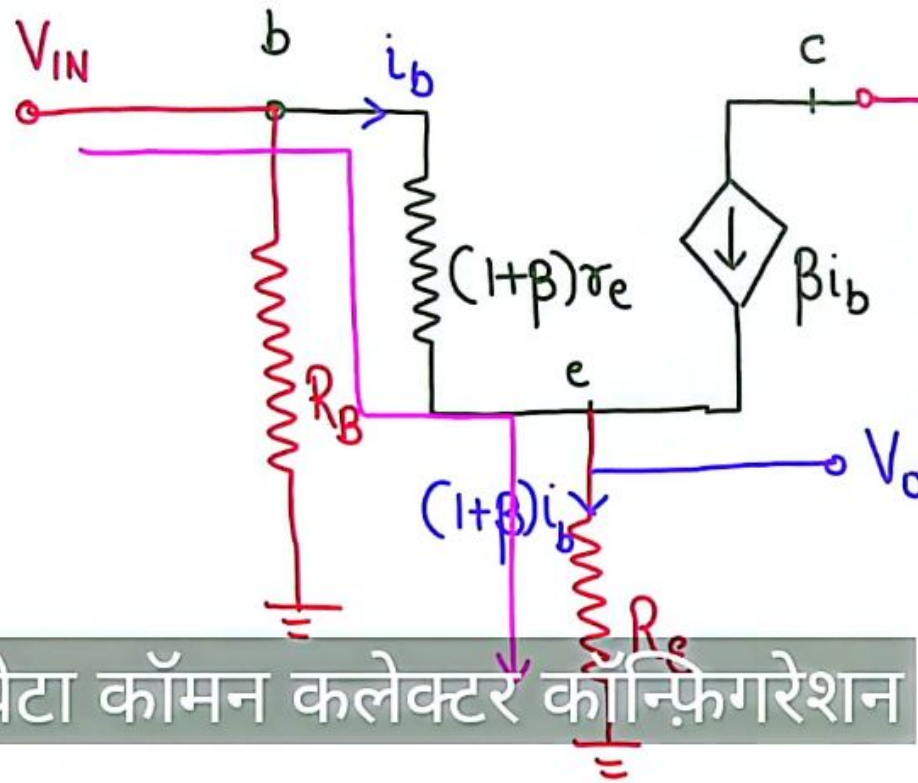
$$\frac{V_O}{V_{IN}} = \frac{R_E}{R_E + r_e} < 1$$

close to 1

वैल्यू या आउटपुट वोल्टेज इसे ऑलमोस्ट इक्वल



Voltage Gain A_V



क्योंकि बेटा कॉमन कलेक्टर कॉन्फिगरेशन में करंट

$$V_{IN} - i_b(1+\beta)r_e - (1+\beta)i_b R_E = 0$$

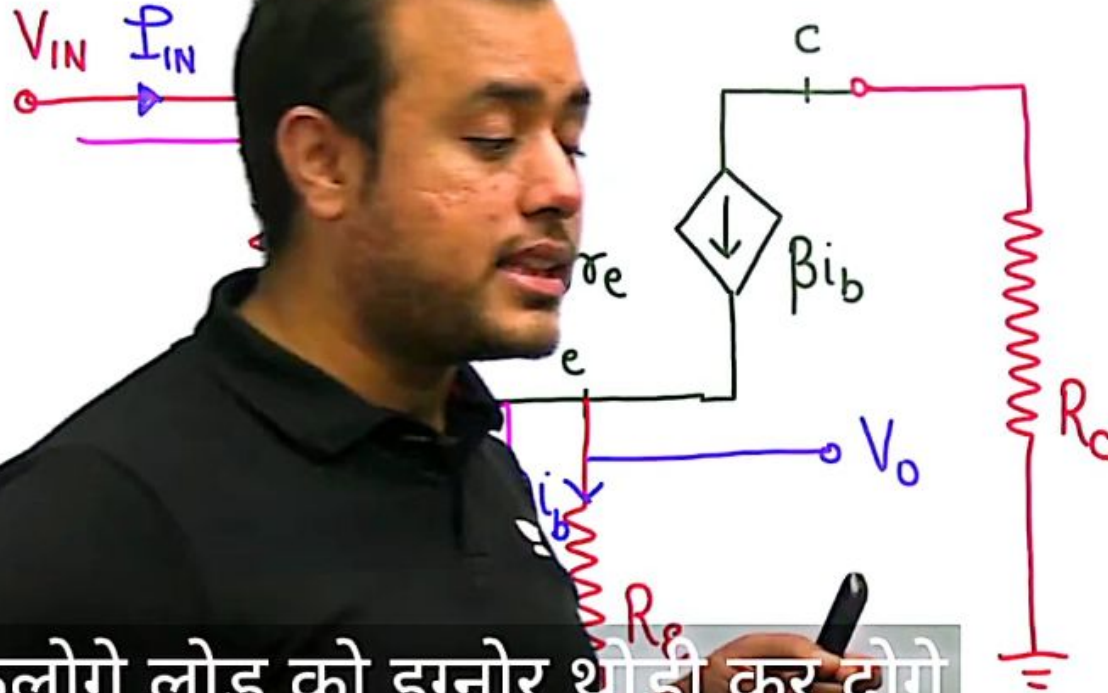
$$\frac{V_{IN}}{(1+\beta)(r_e + R_E)}$$

$$\frac{(1+\beta)i_b R_E}{(1+\beta)(r_e + R_E)} = \frac{R_E}{r_e + R_E} V_{IN}$$



Voltage Gain A_V

- Close to 1 so V_o follows V_{IN}
- 0° shift



जब निकलोगे लोड को इग्नोर थोड़ी कर दोगे
मैंने क्या

$$V_{IN} - i_b(1+\beta)r_e - (1+\beta)i_b R_E = 0$$

$$i_b = \frac{V_{IN}}{(1+\beta)(r_e + R_E)}$$

$$\begin{aligned} V_o &= (1+\beta)i_b R_E \\ &= \frac{(1+\beta)R_E V_{IN}}{(1+\beta)(r_e + R_E)} \end{aligned}$$

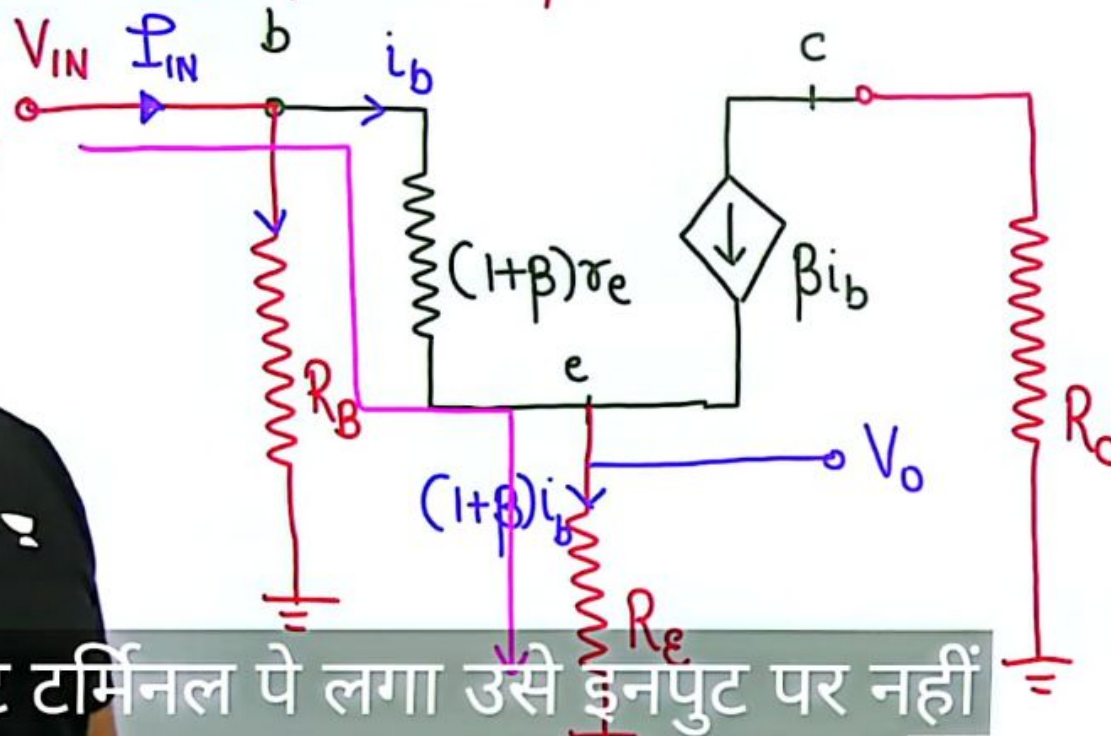
$$\frac{V_o}{V_{IN}} = \frac{R_E}{R_E + r_e} < 1$$

close to 1



Voltage Gain A_V

- Close to 1 so V_o follows V_{IN}
- 0° phase shift



आउटपुट टर्मिनल पे लगा उसे इनपुट पर नहीं आना चाहिए ये तो नहीं थ्योरी हो गई है ना

$$V_{IN} - i_b(1+\beta)r_e - (1+\beta)i_bR_E = 0$$

$$i_b = \frac{V_{IN}}{(1+\beta)(r_e + R_E)}$$

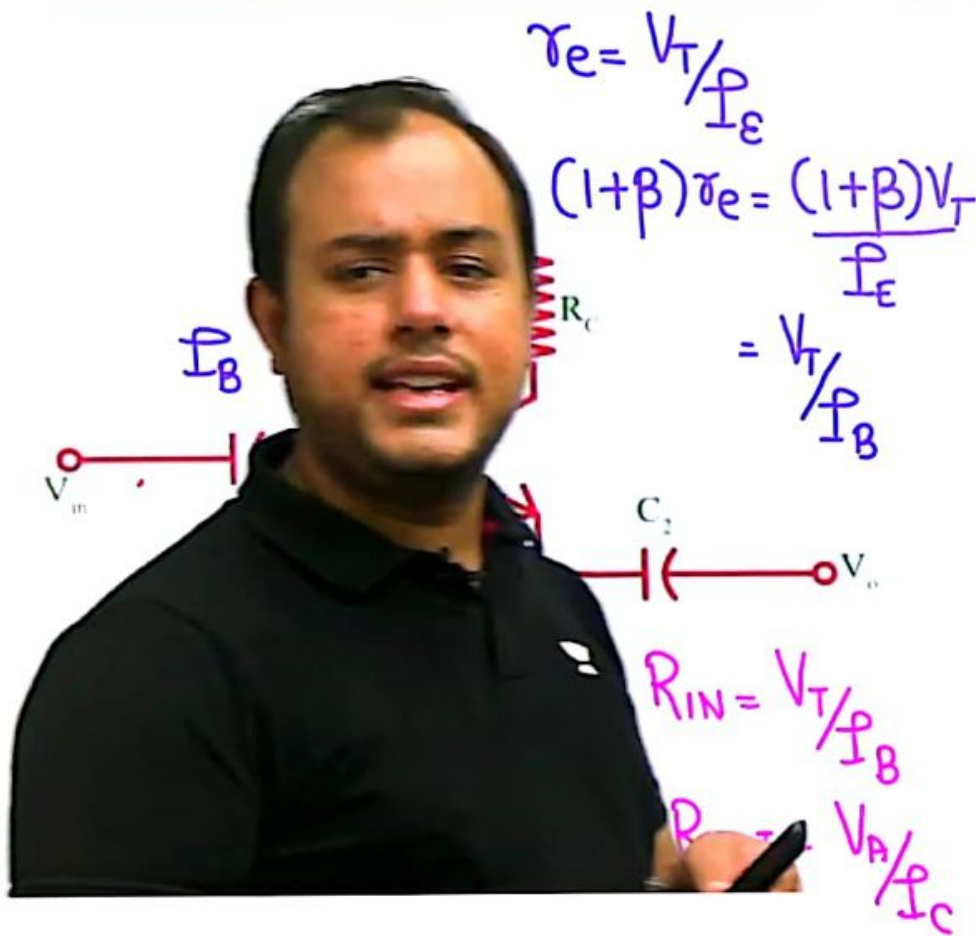
$$\begin{aligned} V_o &= (1+\beta)i_b R_E \\ &= \frac{(1+\beta)R_E V_{IN}}{(1+\beta)(r_e + R_E)} \end{aligned}$$

$$\frac{V_o}{V_{IN}} = \frac{R_E}{R_E + r_e} < 1$$

close to 1



Emitter follower configuration



• o/p is taken from emitter terminal & hence called emitter follower.

• Common Collector Configuration
dc analysis: $C \rightarrow \text{open}$

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - V_{BE} = I_B [R_B + (1 + \beta) R_E]$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta) R_E}$$



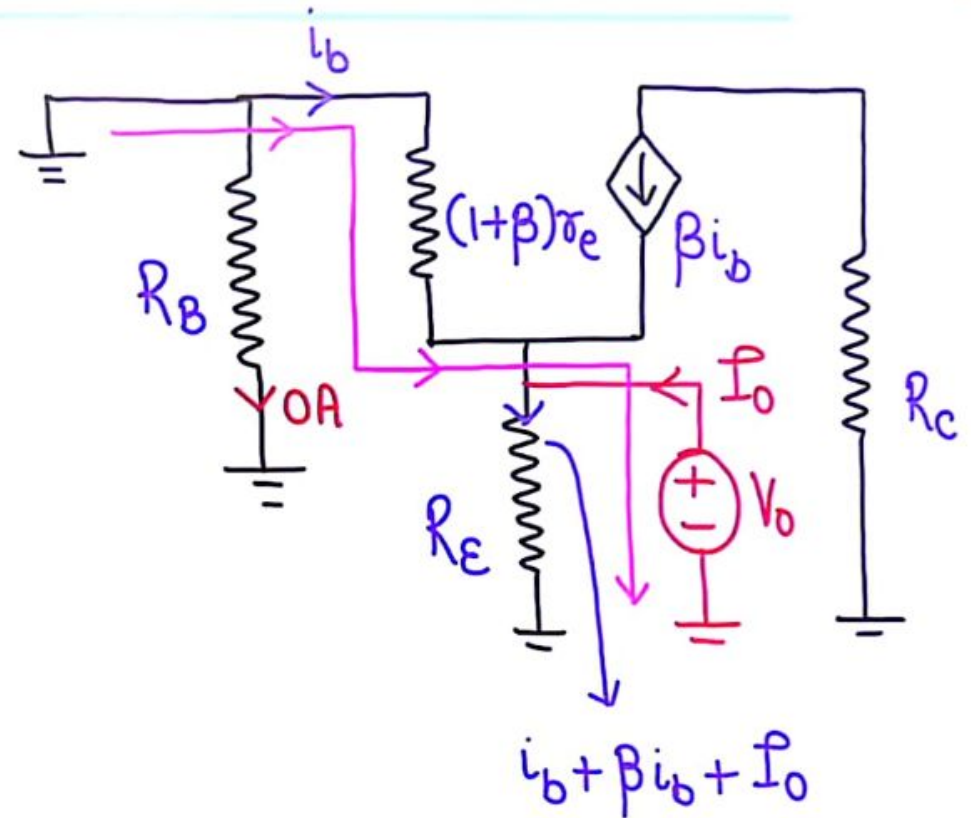
Output Impedance

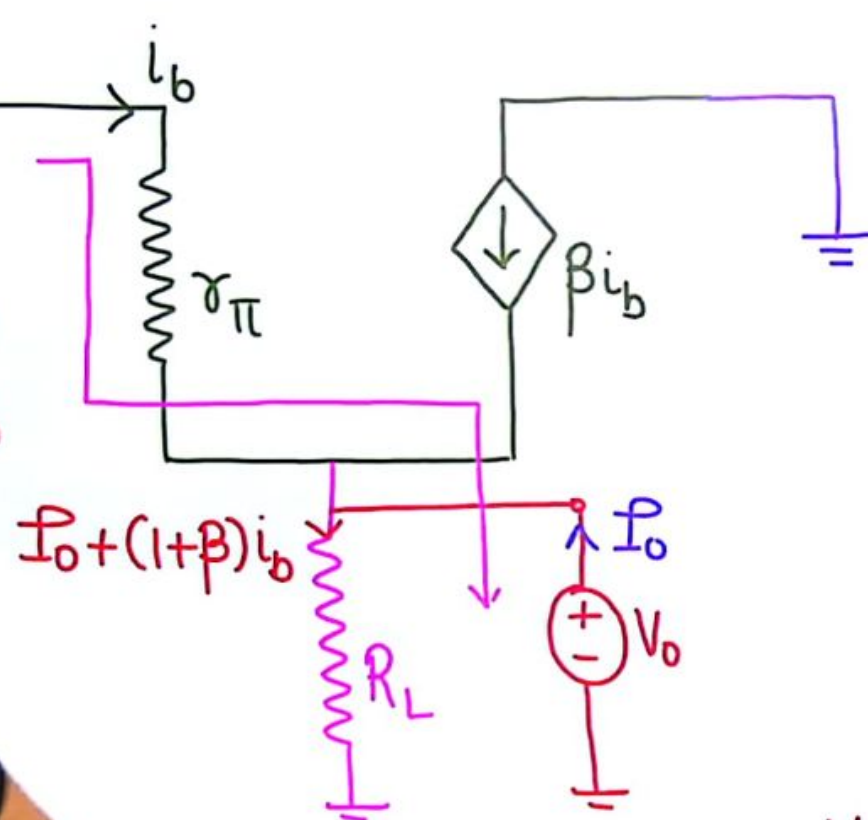
$$R_{OUT} = \frac{V_{OUT}}{I_{OUT}} \bigg|_{V_{IN}=0}$$

$$-i_b(1+\beta)r_e - V_o = 0$$

$$i_b = \frac{-V_o}{(1+\beta)r_e}$$

$$V_o = [(1+\beta)i_b + I_o]R_c$$





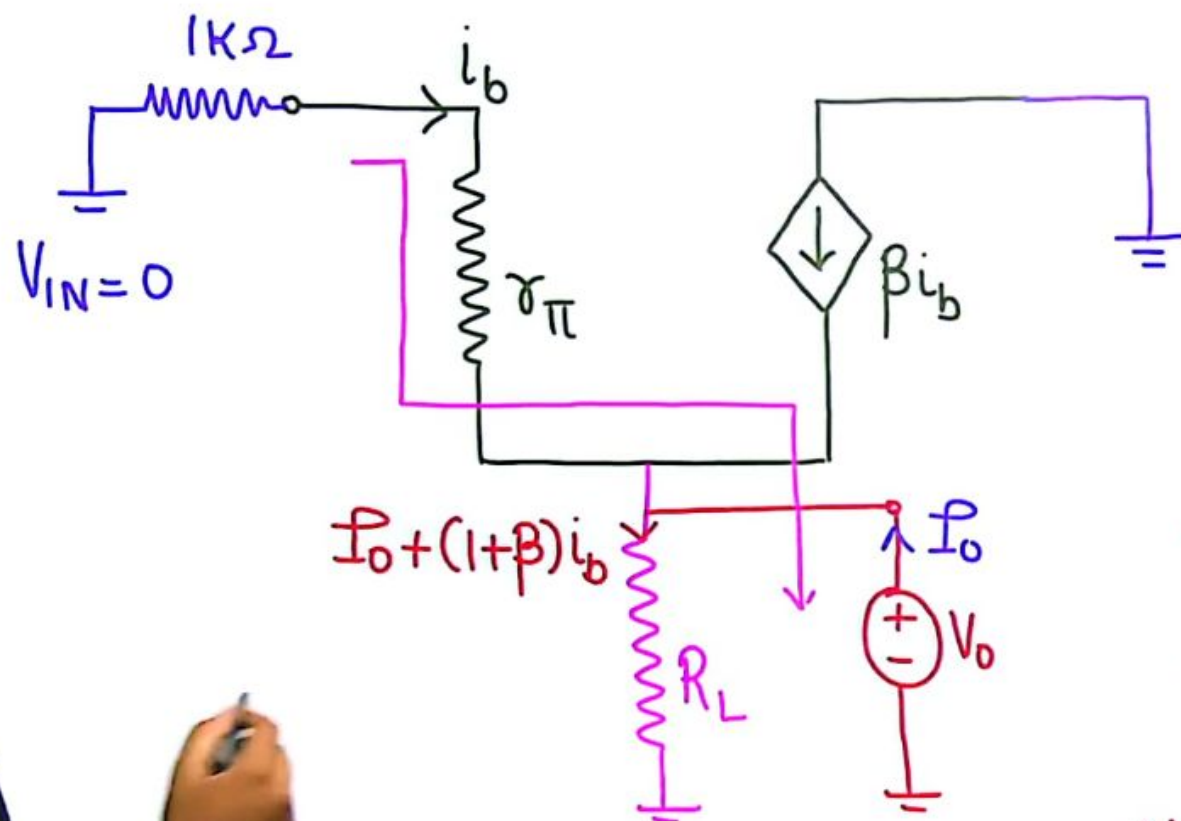
$$0 - i_b(1k) - i_b r_\pi - V_o = 0$$

$$i_b = -\frac{V_o}{2k}$$

$$V_o = R_L (I_o + (1+\beta)i_b)$$

$$V_o = R_L I_o - \frac{100 R_L V_o}{2k}$$

$$\frac{V_o}{I_o} = \frac{R_L}{1 + \frac{100 R_L}{2}}$$



$$0 - i_b(1K) - i_b r_\pi - V_o = 0$$

$$i_b = -\frac{V_o}{2K}$$

$$V_o = R_L (I_o + (1+\beta)i_b)$$

$$V_o = R_L I_o - \frac{100 R_L V_o}{2K}$$

$$\frac{V_o}{I_o} = \frac{R_L}{1 + \frac{100 R_L}{2}}$$



Hybrid Equivalent Model



← input impedance
unit SC
 $V_2 = 0$

← reverse voltage gain
under

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2 = 0}$$

= forward current
gain at SC

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

admittance at

$$h_i (\Omega)$$

$$h_r (\text{unitless})$$

$$h_f (\text{unitless})$$

$$h_o (\Omega)$$

based on configuration we
use subscript in parameters

eg: common emitter

$h_{ie}, h_{re}, h_{fe}, h_{oe}$

$$V_1 = h_i I_1 + h_r V_2 \leftarrow \text{KVL}$$

$$I_2 = h_f I_1 + h_o V_2 \leftarrow \text{KCL}$$

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

= o/p admittance at
OC

- $h_{11} = h_i (\Omega)$
- $h_{12} = h_r$ (unitless)
- $h_{21} = h_f$ (unitless)
- $h_{22} = h_o (\Omega)$

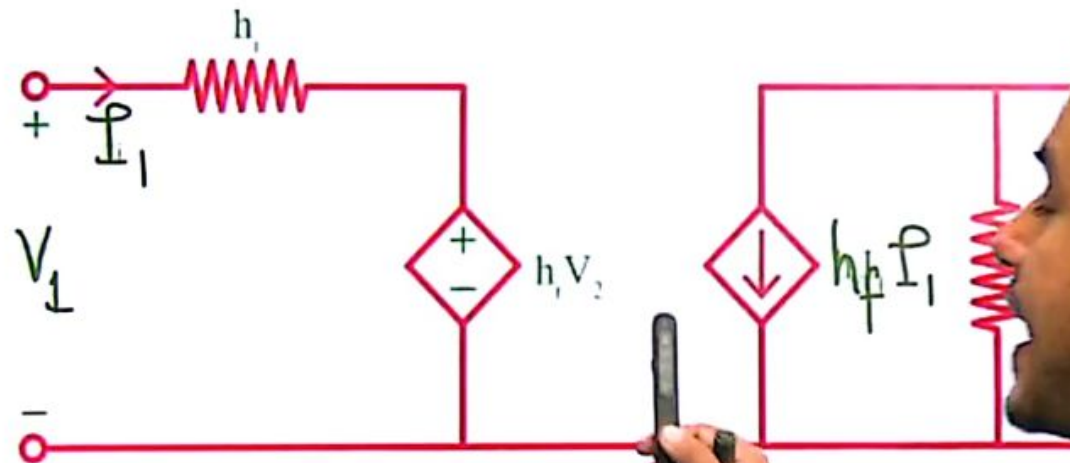
based on configuration we
use subscript in parameters

eg: emitter

hoe

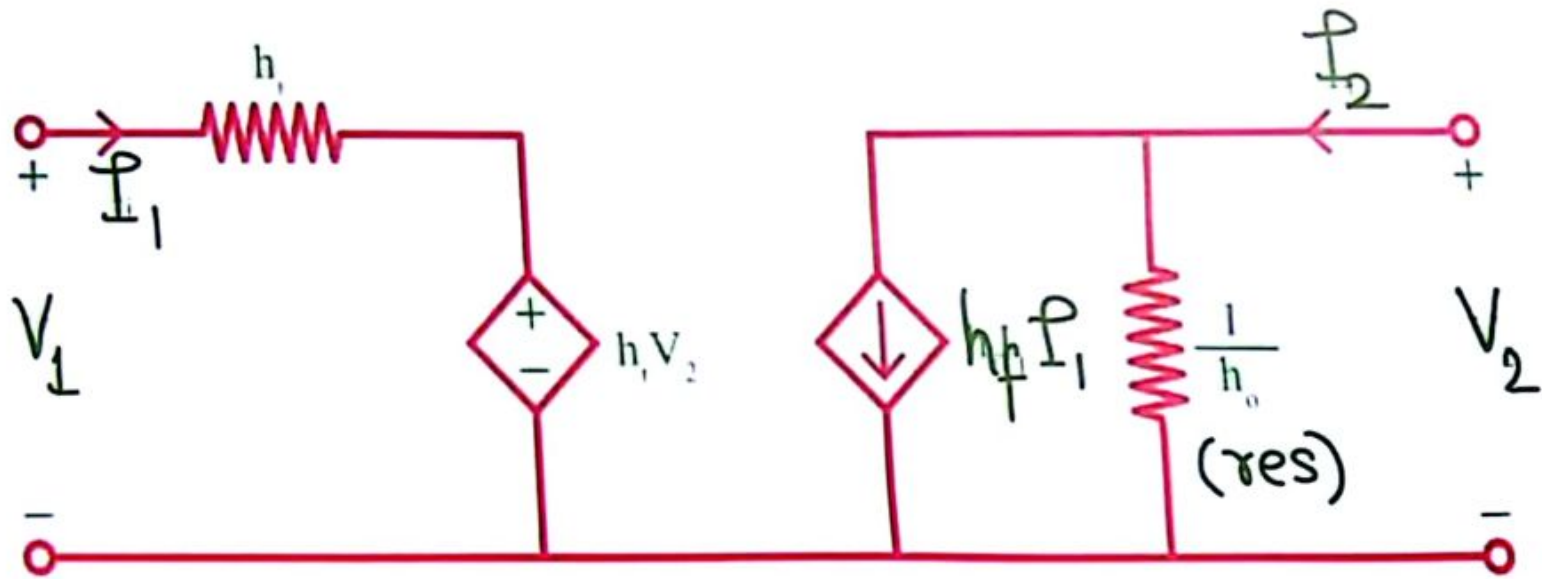
$V_1 = h_i$ ← KVL

$I_2 = h_o V_2$ ← KCL



$$V_1 = h_i I_1 + h_r V_2$$

$$I_2 = \frac{V_2}{\frac{1}{h_f}}$$

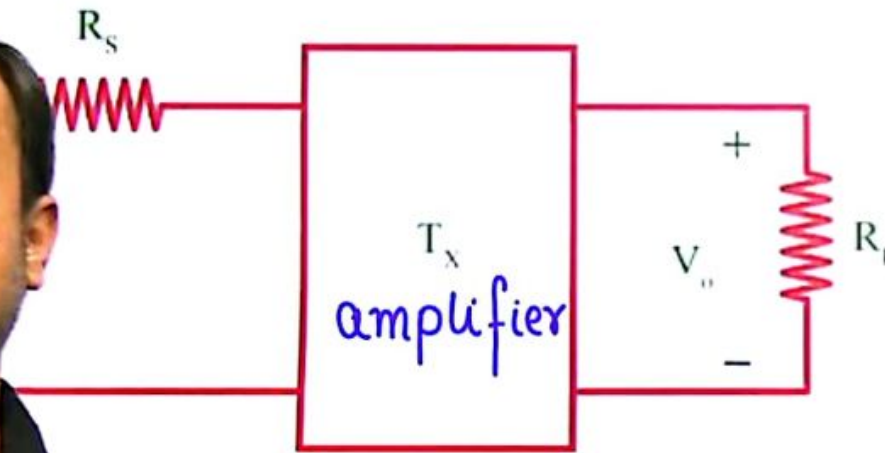


$$V_1 = h_i I_1 + h_r V_2$$

$$I_2 = \frac{V_2}{\frac{1}{h_o}} + h_f I_1$$
$$= h_f I_1 + h_o V_2$$



Transistor Amplifier



- h-model remains same for all configurations
- only subscripts of parameters will change
- position of terminals also change
 - left $\rightarrow$ i/p
 - right $\rightarrow$ o/p
 - bottom $\rightarrow$ common



Transistor Amplifier



• h-model remains same for all configurations

• only subscripts of parameters will change

• position of terminals also change

left $\rightarrow$ i/p

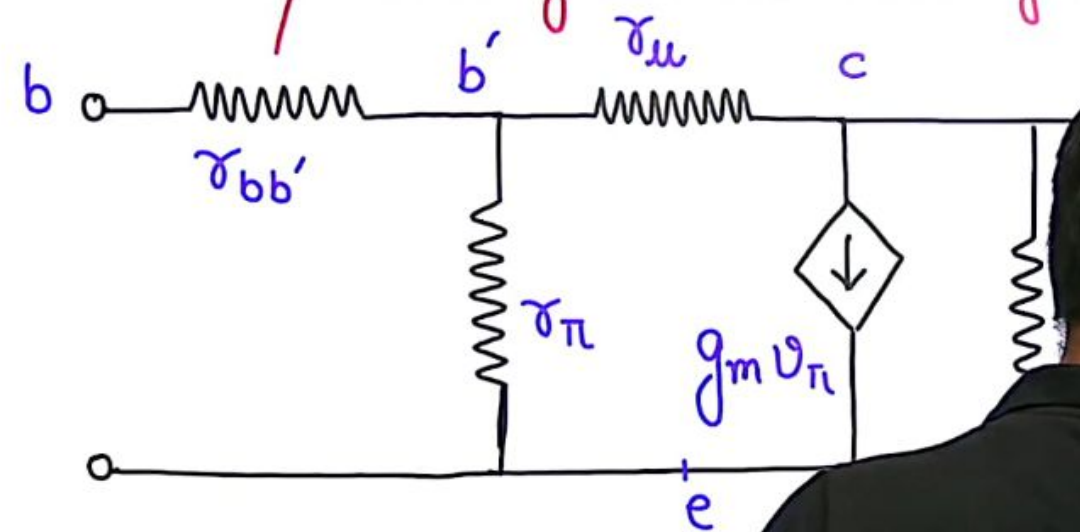
right $\rightarrow$ o/p

bottom $\rightarrow$ common



Hybrid-pi Model

usually considered at high f



r_{μ} : feedback resistance
 due to early effect
 (very high)

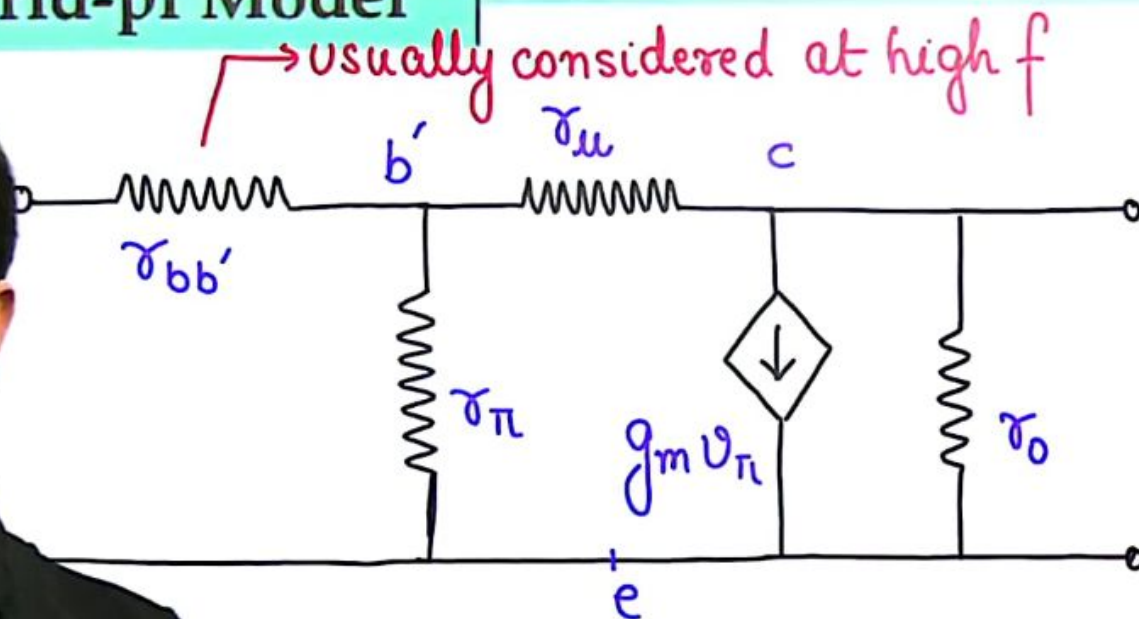
r_{π} : base-emitter res.

$r_{bb'}$: base spreading

↳ resistance
 external to
 ↳ base



Hybrid-pi Model



r_{μ} : feedback resistance
due to early effect
(very high)

r_{π} : base-emitter res.
 $= V_T / I_B$

base spreading resistance

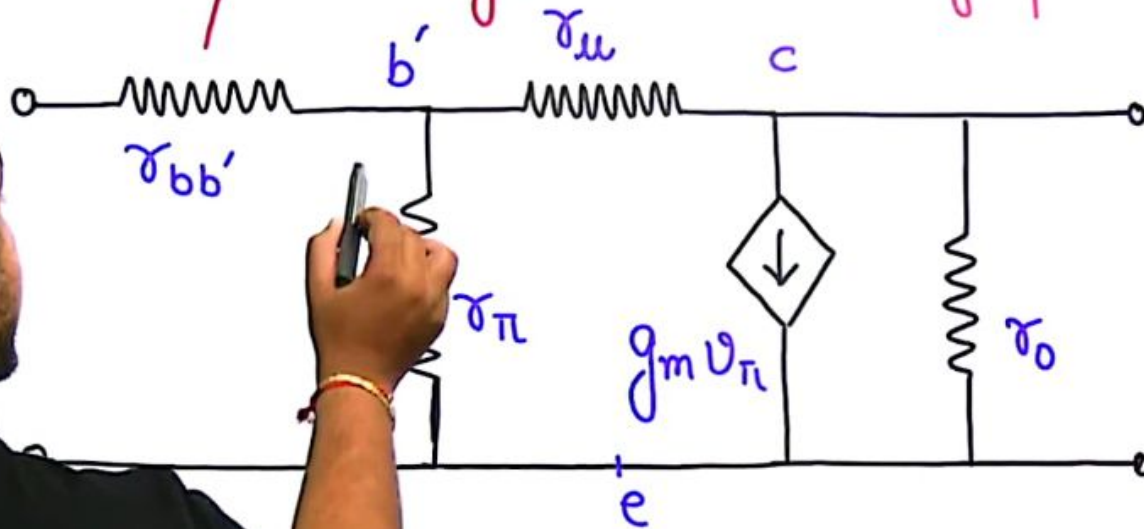
→ resistance b/w internal &
external base terminal

↳ base area is smallest & $r = \text{highest}$



Hybrid-pi Model

→ usually considered at high f



r_{μ} : feedback resistance
due to early effect
(very high)

r_{π} : base-emitter res.
 $= V_T / I_B = (1 + \beta) r_e$

spreading resistance
resistance b/w internal &
external base terminal

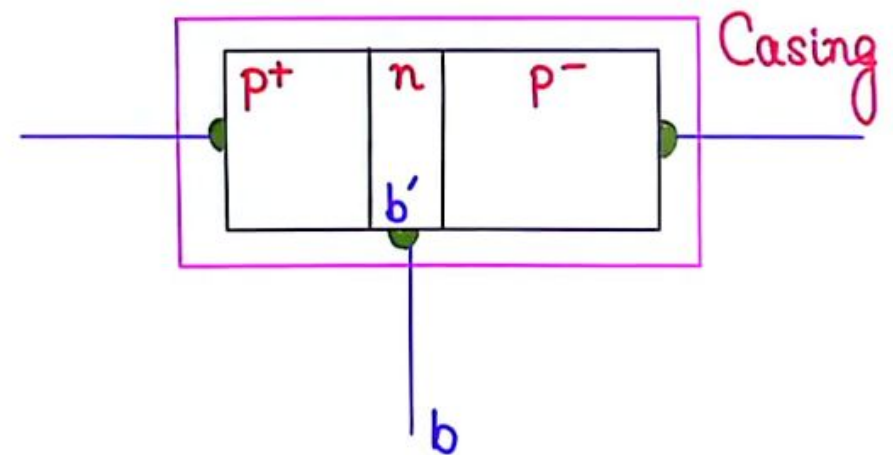
→ base area is smallest & $r = \text{highest}$

g_m - transconductance

lector current

• BJT is modeled as VCCS instead of CCCS.

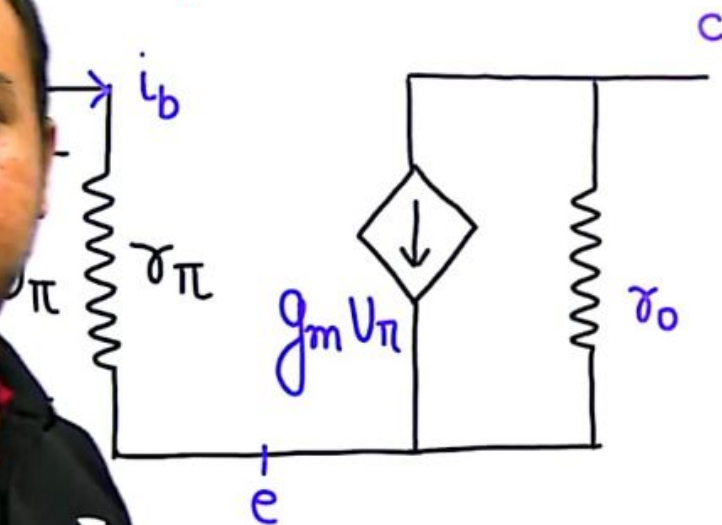
• model remains same for npn & pnp $\bar{I}_x$.





Simplified Hybrid-pi Model

o ignore $r_{bb'}$ & r_u



$$r_{\pi} = V_T / I_B ; g_m = I_C / V_T$$

$$g_m r_{\pi} = I_C / I_B = \beta$$

current source

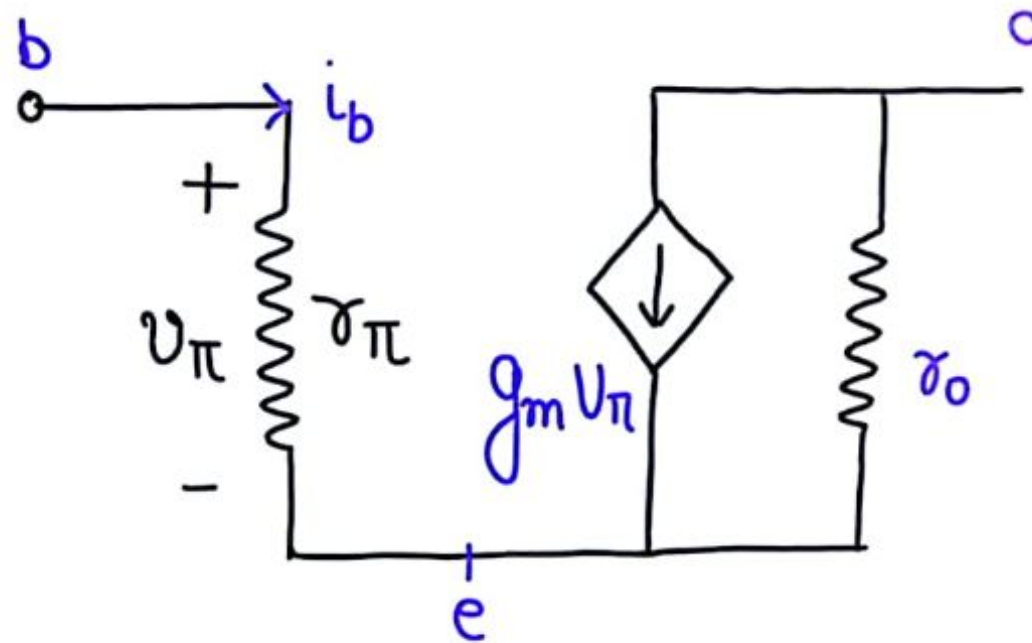
$$\begin{aligned} &= g_m v_{\pi} = g_m r_{\pi} i_b \\ &= \beta i_b \end{aligned}$$

$g_m r_{\pi} i_b$



Simplified Hybrid-pi Model

o ignore $r_{bb'}$ & r_u



$$v_{\pi} = i_b r_{\pi}$$

$$g_m v_{\pi} = g_m r_{\pi} i_b$$

$$r_{\pi} = V_T / I_E$$

$$g_m r_{\pi} = \beta$$

current source

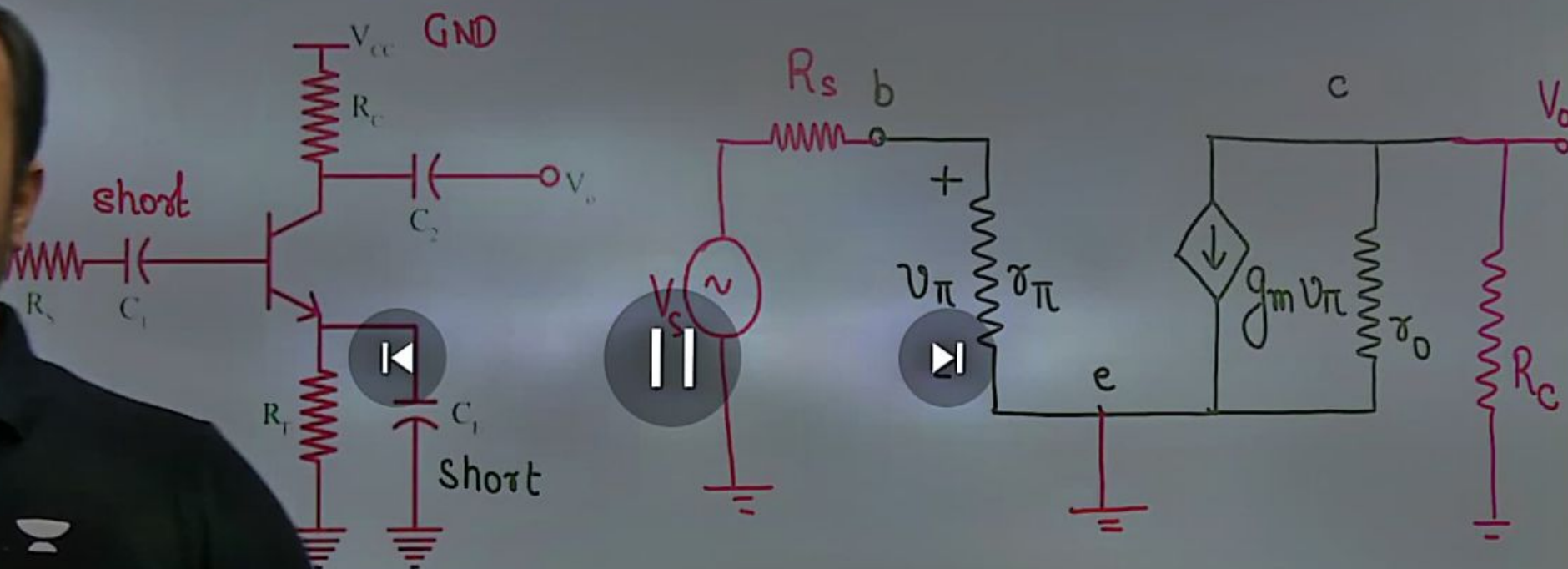
$$= g_m v_{\pi} = g_m$$

$$= \beta i_b$$

Lakshya Batch 2.0 - Analog Electronics | AC Models of BJT Part - 04 | Ankit

Kreatryx GATE - EE, ECE & IN by Unacademy

(1) Common Emitter Amplifier With Bypass Capacitor



/ 1:20:36 (1.75x)

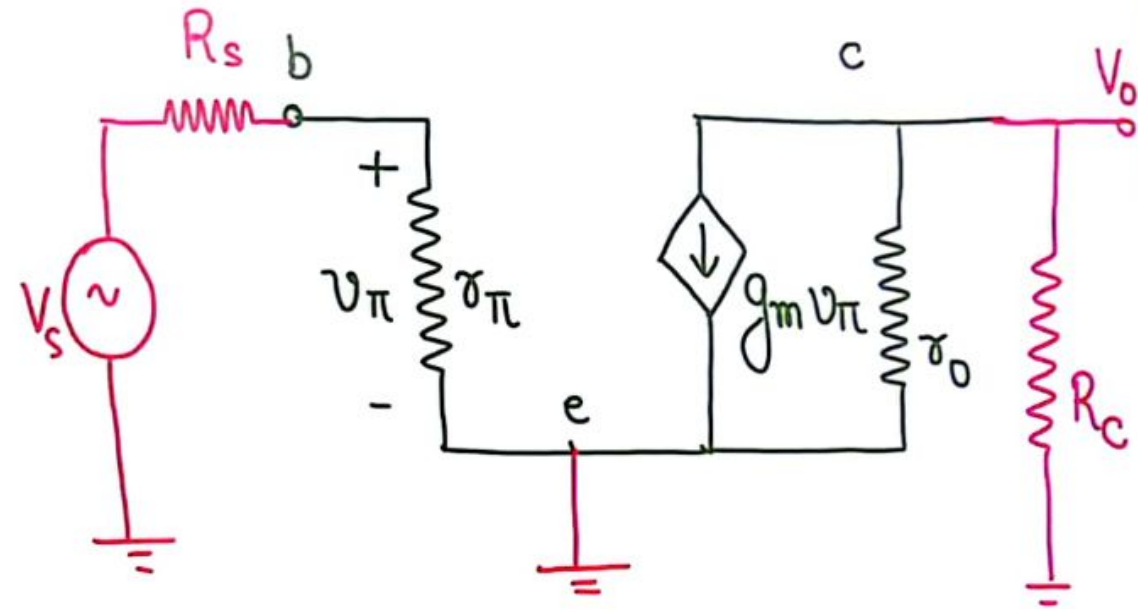
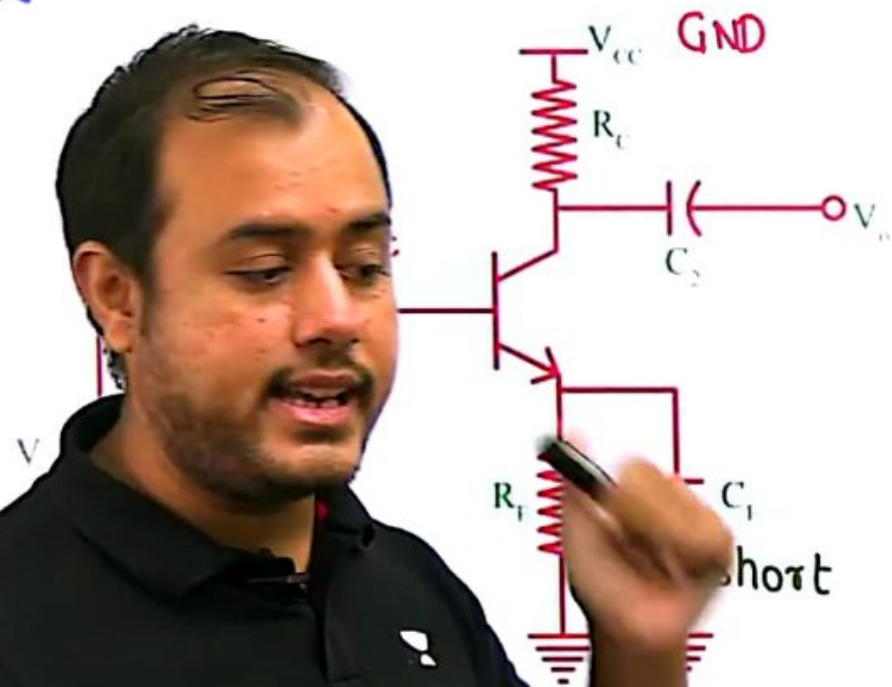


More videos
Tap or swipe up to see all





(1) Common Emitter Amplifier With Bypass Capacitor





Voltage Gain A_v

$$V_o = -g_m v_{\pi} (r_o \parallel R_c)$$

$$V_{IN} = v_{\pi}$$

$$A_v = \frac{V_o}{V_{IN}} = -g_m (r_o \parallel R_c) \approx -g_m R_c$$

Input Resistance

$$R_{IN} = r_{\pi}$$



Voltage Gain A_v

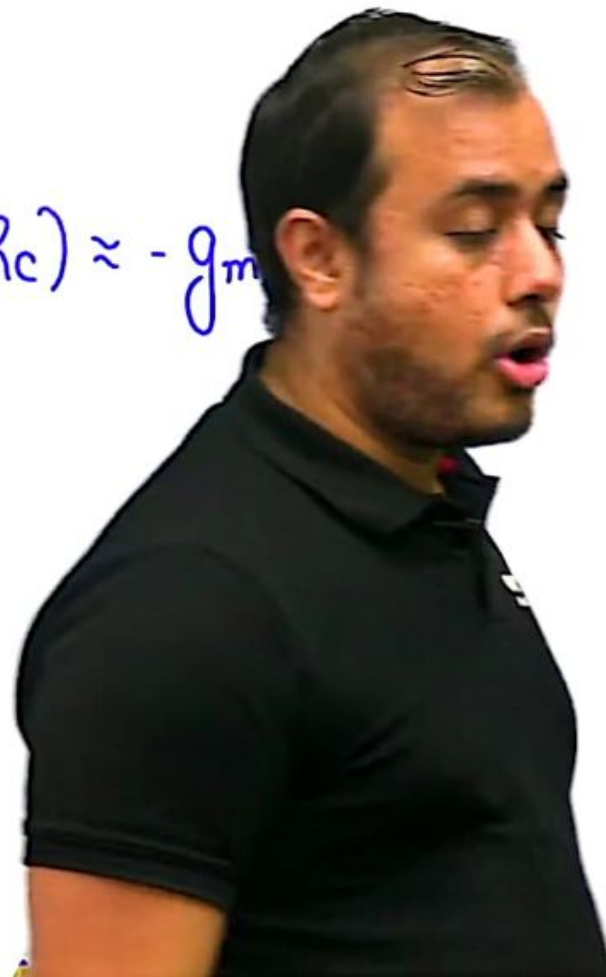
$$V_o = -g_m v_{\pi} (r_o \parallel R_c)$$

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Input Resistance

$$R_{IN} = r_{\pi}$$





Overall voltage Gain A_{vs}

$$V_o = -g_m v_\pi (\tau_o \parallel R_c)$$

$$v_\pi = V_s \frac{\tau_\pi}{\tau_\pi + R_s}$$

$$\frac{V_o}{V_s} = -g_m \frac{\tau_\pi}{\tau_\pi + R_s} (\tau_o \parallel R_c)$$

Output Resistance R_o

$$R_o = \tau_o \parallel R_c$$

11:58



Overall voltage Gain A_{vs}

$$V_o = -g_m v_\pi (\tau_o \parallel R_c)$$

$$v_\pi = V_s \frac{\tau_\pi}{\tau_\pi + R_s}$$

$$\frac{V_o}{V_s} = -g_m \frac{\tau_\pi}{\tau_\pi + R_s} (\tau_o \parallel R_c)$$

Output Resistance R_o

$$R_{out} = \tau_o \parallel R_c$$

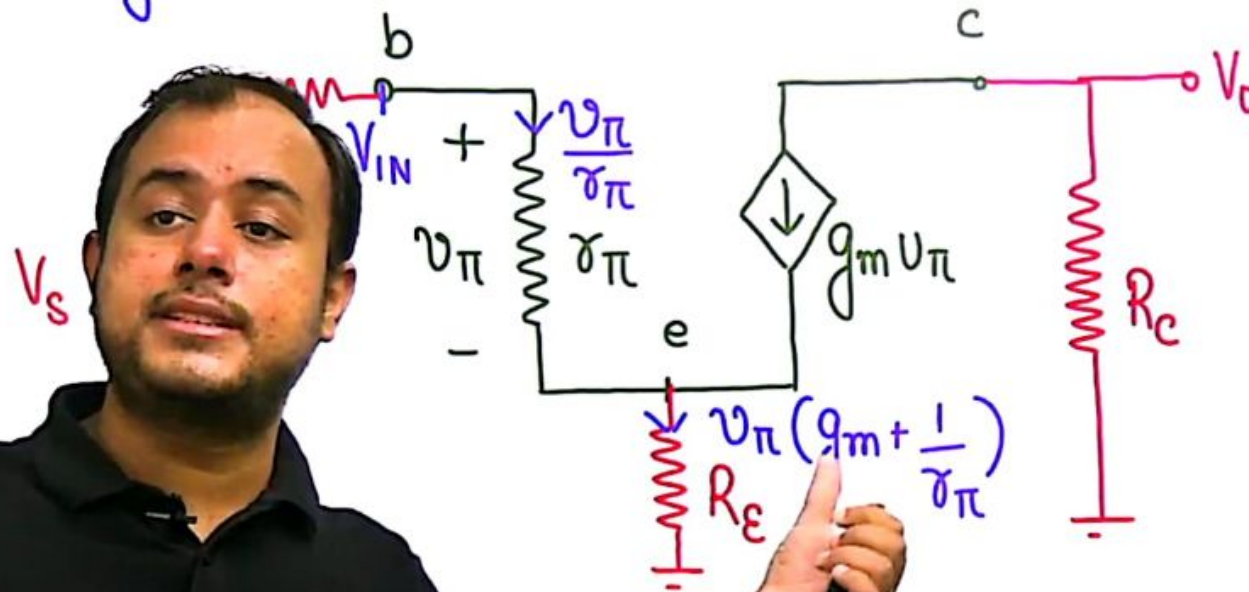


11:58



(2) Common Emitter Amplifier Without Bypass Capacitor

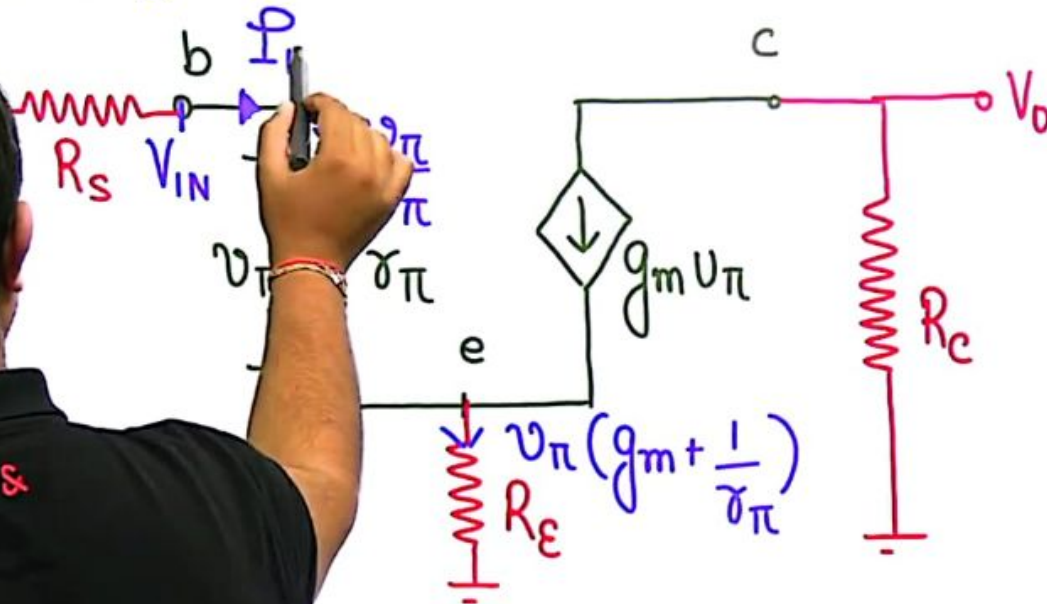
ignore r_o





(2) Common Emitter Amplifier Without Bypass Capacitor

ignore r_o



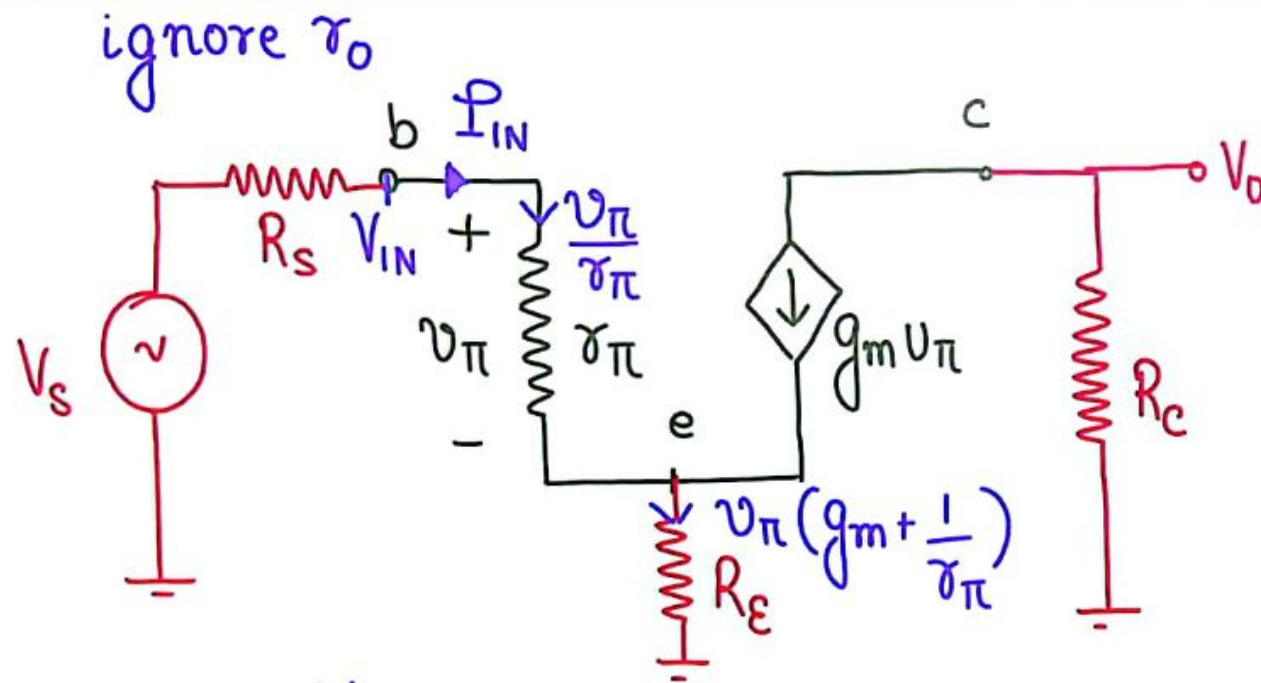
$$V_O = -g_m v_{\pi} R_C$$

$$v_{\pi} + v_{\pi} \left(g_m + \frac{1}{r_{\pi}} \right) R_E$$

$$v_{\pi} \left(1 + g_m R_E + \frac{R_E}{r_{\pi}} \right)$$



(2) Common Emitter Amplifier Without Bypass Capacitor



$$V_{IN} = v_{\pi} + v_{\pi} \left(g_m + \frac{1}{r_{\pi}} \right) R_E$$

$$V_{IN} = v_{\pi} \left(1 + g_m R_E + \frac{R_E}{r_{\pi}} \right)$$





Input Resistance R_i

$$\begin{aligned} R_{IN} &= V_{IN} / I_{IN} \\ &= \frac{v_{\pi} (1 + g_m R_E + R_E / r_{\pi})}{v_{\pi} / r_{\pi}} \\ &= r_{\pi} + R_E + g_m r_{\pi} R_E \\ g_m r_{\pi} &= \beta \end{aligned}$$



Input Resistance R_i

$$R_{IN} = \frac{V_{IN}}{I_{IN}}$$
$$= \frac{v_{\pi} (1 + g_m R_E + R_E / r_{\pi})}{v_{\pi} / r_{\pi}}$$

$$= r_{\pi} + R_E + g_m r_{\pi} R_E$$

$$g_m r_{\pi} = \beta$$

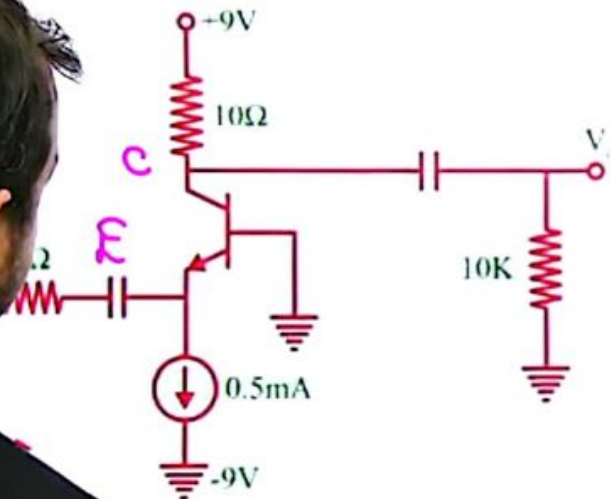
$$R_{IN} = r_{\pi} + (1 + \beta) R_E$$





Question-02

Given $\beta = 100$, Find R_{in} , $\frac{V_o}{V_s}$ (voltage gain)

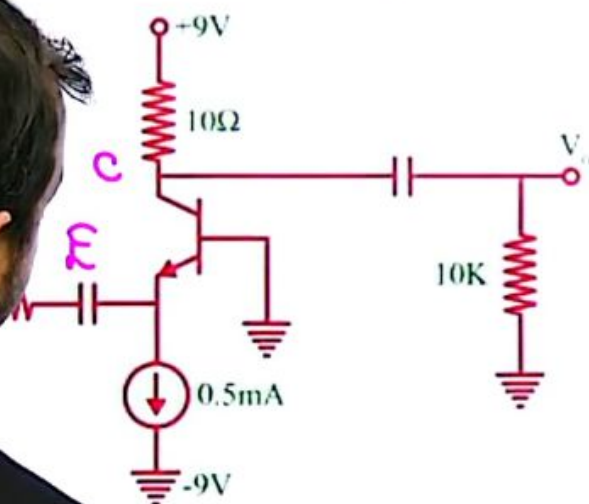


Common



Question-02

Given $\beta = 100$, Find R_{in} , $\frac{V_o}{V_s}$ (voltage gain)



Common base



Frequency Response of BJT

Frequency Response

Signal
generator

Amplifier



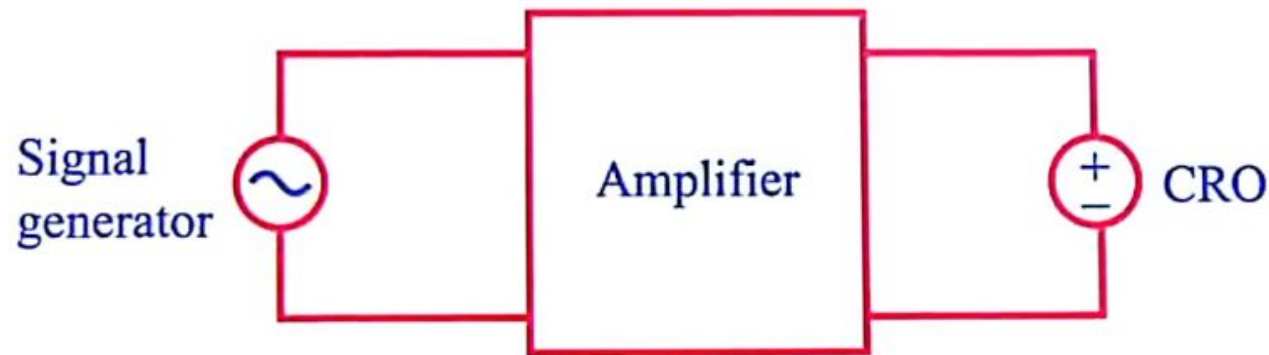
CRO

• when the frequency of input signal varies then gain of amplifier also varies. The variation of gain with f is called as **Frequency Response**.



Frequency Response of BJT

Frequency Response



◦ variation in gain is due to capacitances

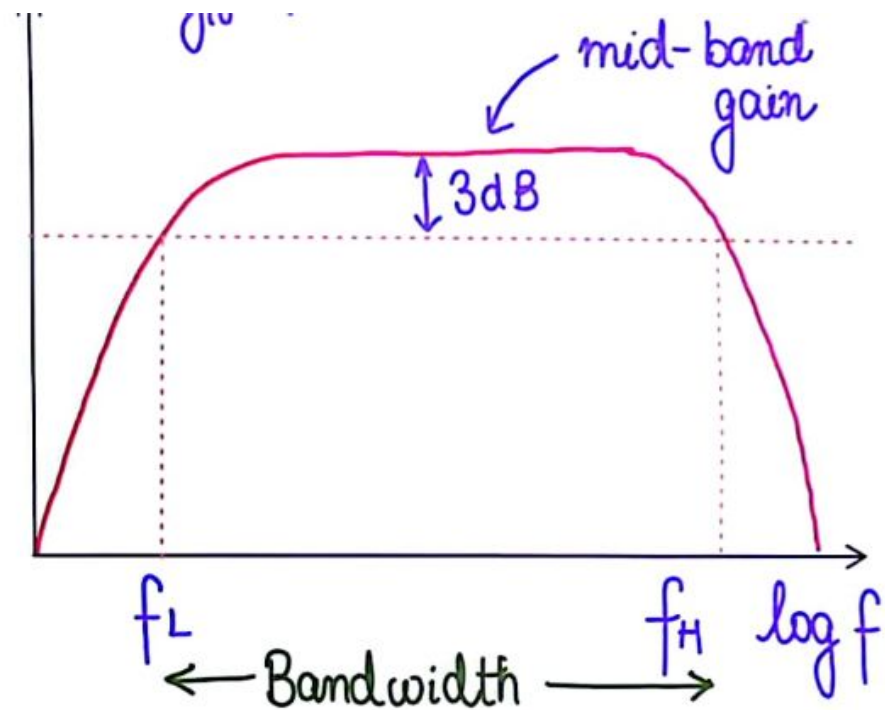
→ Coupling capacitance, By pass cap

→ stray cap, internal cap (diff)

◦ when the frequency of input signal increases then gain of amplifier also varies. The variation of gain with frequency is called

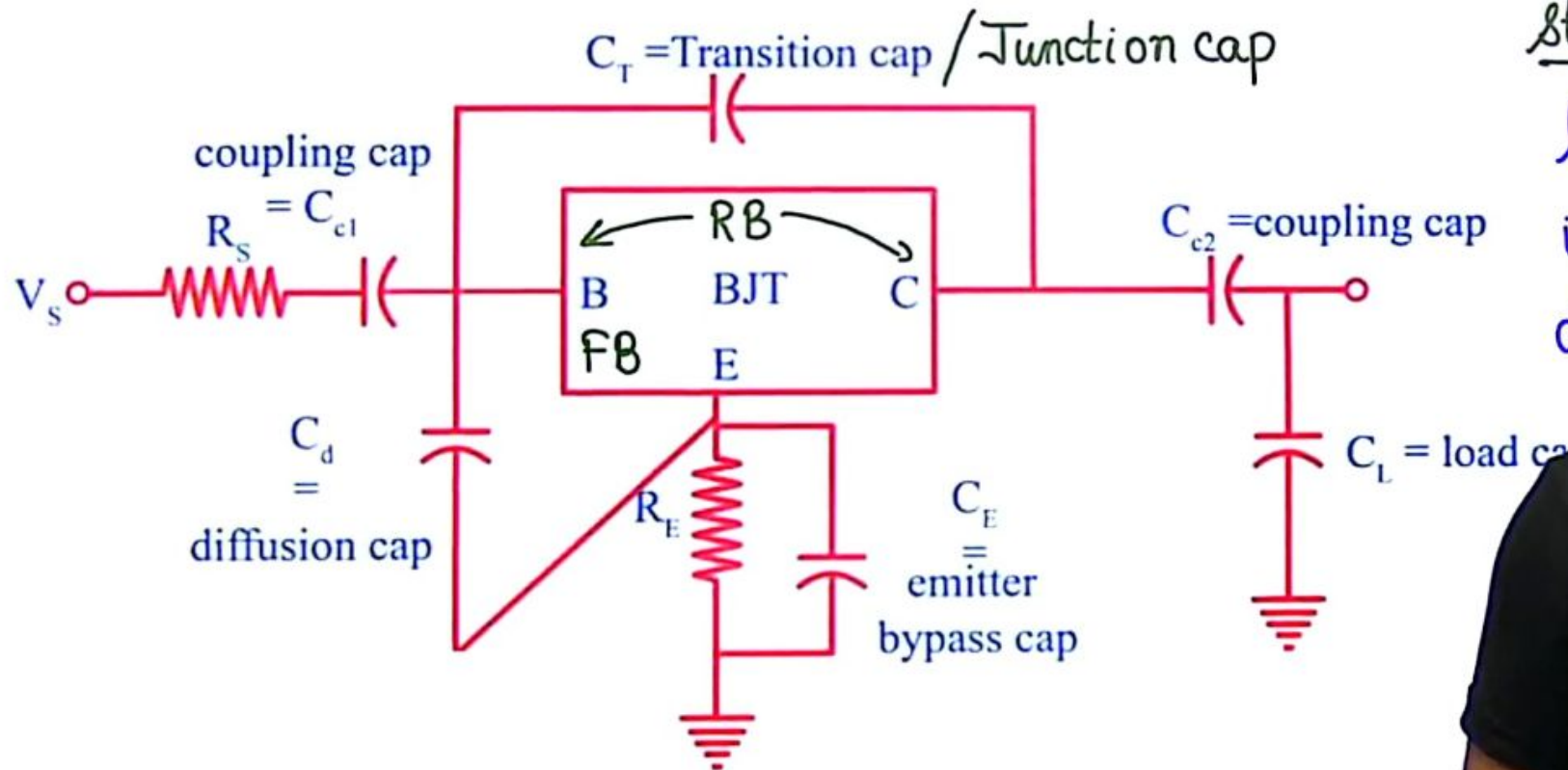
f	V_o	$A_v = V_o/V_i$	$20 \log_{10} A_v$

by means of signal generator
are varying input freq.
amplitude of
ant; usually 10mV (p-p)





Frequency Response of BJT

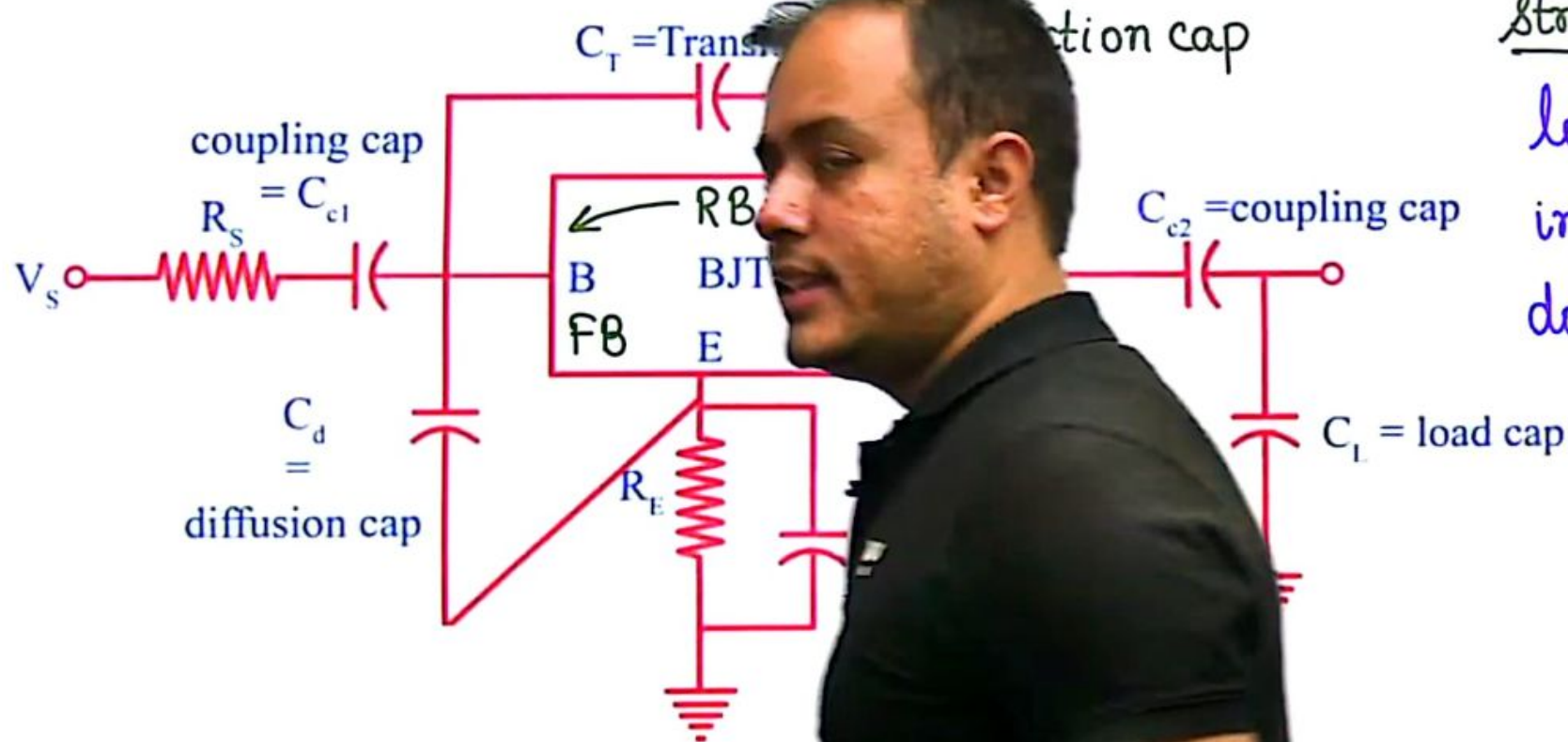


Stray cap
located
in device
don't know





Frequency Response of BJT



Stray capacitance
located somewhere
in device but we
don't know
location.

Frequency Range	Coupling and Bypass Capacitor	Load and parasitic Capacitor
	C_{C1}, C_{C2}, C_E [μF] (low f) high	C_d, C_T, C_L [pF] (high f) low
Low frequency	Consider $X_C = 1/\omega C$ $\omega = \text{low}$ $C = \text{high}$ $X_C = \text{finite}$	$X_C = 1/\omega C$ $\omega = \text{low}$ $C = \text{high}$
Mid band	short $X_C = \text{low}$	
High frequency	short $\omega = \text{high}$ $C = \text{high}$ $X_C = \text{low}$	$X_C = \text{finite}$

Frequency Range	Coupling and Bypass Capacitor	Load and parasitic Capacitor
	C_{C1}, C_{C2}, C_E [μF] (low f) high	C_d, C_T, C_L [pF] (high f) low
Low frequency	Consider $X_C = \frac{1}{\omega C}$ $\omega = \text{low}$ $C = \text{high}$ $X_C = \text{finite}$	open $X_C = \frac{1}{\omega C}$ $\omega = \text{low}$ $C = \text{low}$ $X_C = \text{high}$
Mid band	short $X_C = \text{low}$	open $X_C = \text{high}$
High frequency	short $\omega = \text{high}$ $C = \text{high}$ $X_C = \text{low}$	Consider $\omega = \text{high}$ $C = \text{low}$ $X_C = \text{finite}$

Frequency Range	Coupling and Bypass Capacitor	Load and parasitic Capacitor
	C_{C1}, C_{C2}, C_E [μF] (low f) high	C_d, C_T, C_L [pF] (high f) low
Low	Consider $X_C = \frac{1}{\omega C}$ $\omega = \text{low}$ $C = \text{high}$ $X_C = \text{finite}$	open $X_C = \frac{1}{\omega C}$ $\omega = \text{low}$ $C = \text{low}$ $X_C = \text{high}$
	short $X_C = \text{low}$	open $X_C = \text{high}$
	short $\omega = \text{high}$ $C = \text{high}$ $X_C = \text{low}$	Consider $\omega = \text{high}$ $C = \text{low}$ $X_C = \text{finite}$